BEFORE THE NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	)	**Case 23-G-0225**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service	)	**Case 23-G-0226**

DIRECT TESTIMONY OF SONAL JESSEL
ON BEHALF OF WE ACT FOR ENVIRONMENTAL JUSTICE

September 1, 2023

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I. INTRODUCTIONS, QUALIFICATIONS, AND RECOMMENDATIONS

Q. Please state your name and business address.

A. My name is Sonal Jessel and I am the Director of Policy at WE ACT for Environmental Justice.

Q. Please describe your organization.

A. WE ACT for Environmental Justice is a community-based organization that, since 1988, has been building healthy communities by ensuring that people of color and/or low-income residents participate meaningfully in the creation of sound and fair environmental health and protection policies and practices. Today, the organization has grown to over 35 staff members in New York City and Washington, D.C., and is considered an active and respected participant in the national Environmental Justice Movement.

Q. Please summarize your background and relevant experience.

A. I have a master's in public health from Columbia University's Mailman School of Public Health. My concentration for the degree was in Climate and Health. I conducted research in energy insecurity as a Graduate Assistant with Dr. Diana Hernández while at Columbia University. As the Director of Policy at WE ACT for Environmental Justice, I am responsible for advancing the organization's policy agenda at the local and state levels and am responsible for the enactment of many policies. I am an appointee to the New York State Climate Impacts Assessment, Health and Safety Technical Workgroup. Additionally, I am a member of the New York State Climate Justice Working Group, tasked with defining "disadvantaged communities" ("DACs") for the Climate Leadership and Community Protection Act ("CLCPA").

Q. What is the Climate Justice Working Group and what has been your role?

A. The Climate Justice Working Group ("CJWG") is a working group mandated by the CLCPA to define "disadvantaged communities." The CJWG is made up of representatives from state agencies and environmental justice communities. My role on the CJWG is as a representative of an environmental justice community. I attend meetings and advise on the development of the criteria to define DACs. I also contributed suggestions to the Climate Action Council on the Scoping Plan, gave feedback on the State's Community Air Monitoring project, and advised all Climate Action Council advisory panels on the development of their recommendations. The CJWG is mandated to meet at least once annually in perpetuity to adjust the DAC criteria as needed.

Q. Have you testified in a previous rate case before the New York Public Service Commission?

A. Yes. I testified before the New York Public Service Commission (the "Commission") in the 2022 Consolidated Edison rate case, the Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service (Case 22-E-0064) and for Gas Service (Case 22-G-0065).

Q. On whose behalf are you testifying in this case?

A. I am testifying on behalf of WE ACT for Environmental Justice.

Q. What is the purpose of your testimony?

A. The purpose of my testimony is 1) to discuss the development of the DACs definition and maps as outlined by the CLCPA, 2) to discuss the impacts of fossil fuel combustion, alternative fuel combustion, and energy burdens, particularly in DACs, and 3) to recommend changes to KEDNY and KEDLI's (the "Companies'") rate proposals that

would help reduce disproportionate burdens on disadvantaged communities and ensure that at least 35% of the benefits of energy efficiency and clean energy programs accrue to DACs as mandated by the CLCPA.

The purpose of my testimony is also to discuss KEDNY and KEDLI's ratepayer-funded advertising and political activity and to recommend changes to the Companies' rate proposals to ensure that the utilities are not improperly passing political expenditures onto ratepayers.

Q. What information did you review in preparing this testimony?

A. Research articles, reports, grey literature, discovery responses, and media.

Q. Are you sponsoring any exhibits with your testimony?

A. Yes. I am attaching the following exhibits:

- **Exhibit 1:** Jeremy Hoffman et al., *The Effects of Historical Housing Policies on Resident Exposure to Intra-Urban Heat: A Study of 108 US Urban Areas*, 8 Climate 12 (2020).
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Q. Please summarize your conclusions and recommendations.

A. I begin by defining environmental justice and describing the significance of the terminology of "disadvantaged communities" in the CLCPA. I then outline how the Companies are subject to CLCPA requirements. Then, I describe how DACs are disproportionately burdened by environmental and public health burdens from the Companies' fossil fuel infrastructure and combustion in homes and would be disproportionately burdened by RNG and hydrogen infrastructure and combustion in homes. I particularly focus on energy burden and air pollution in homes as central issues that disproportionately impact these populations. From there, I describe how the Companies can reduce their contributions to disproportionate environmental and public health burdens primarily through energy efficiency and electrification. I then provide specific recommendations. The Companies must increase funding to energy efficiency and electrification programs with specific strategies to ensure these investments reduce health and energy burdens in DACs. The Companies must direct at least 35–40% of their investments to disadvantaged communities to reduce pollution burden, increase public

education, meet workforce hiring targets, and reduce utility bill rates for low-income customers.

I then discuss the Companies' ratepayer-funded political activities. State and federal law as well as guidance from the Commission prohibit utilities from passing political expenditures onto ratepayers. In April 2022 National Grid released a report—paid for by ratepayers—called "Our clean energy vision," which makes policy recommendations for using alternative combustion fuels to decarbonize New York's gas supply and discourages policymakers from supporting full electrification of the building sector. National Grid has used the report as a political advocacy tool, including in comments to the Climate Action Council on the State's Scoping Plan. The Companies also use the report to promote National Grid's positions on policy issues through social media and in communications with customers for which ratepayers foot the bill. The Companies are also seeking recovery for trade association dues to at least one organization that engages in lobbying, and for other customer communications that promote National Grid's positions on climate policy issues. These costs should be excluded from the Historic Test Year and the Commission should require the Companies to track political expenditures more stringently to ensure that ratepayers are not improperly shouldering those costs.

II. OVERVIEW OF ENVIRONMENTAL JUSTICE

Q. What is environmental justice?

A. The U.S. Department of Energy defines environmental justice as "the fair treatment and meaningful involvement of all people, regardless of race, color, national origin, or

income, with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies."[1]

Q. What is your understanding of fair treatment and meaningful involvement?

A. According to the U.S. Department of Energy,

> Fair treatment means that no population bears a disproportionate share of negative environmental consequences resulting from industrial, municipal, and commercial operations or from the execution of federal, state, and local laws; regulations; and policies. Meaningful involvement requires effective access to decision makers for all, and the ability in all communities to make informed decisions and take positive actions to produce environmental justice for themselves.[2]

An example of an ongoing environmental justice concern arising from a policy that denied communities fair treatment and meaningful involvement is the legacy of redlining. Redlining refers to a series of policies arising in the 1930s that allowed for the segregation of housing stock based on race. These policies allowed governments and private entities to avoid investments in neighborhoods of color. Research demonstrates that, today, formerly redlined neighborhoods have higher extreme heat vulnerability, less green space, and more toxic facilities, warehouses, highways, and infrastructure.[3] Heat vulnerability is defined by the New York State Department of Health as "how likely a person is to be injured or harmed during periods of hot weather."[4] Research has shown that heat vulnerability is tied to sociodemographic and built environment characteristics.

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[1] *See* Off. of Legacy Mgmt., *What is Environmental Justice?*, U.S. Dep't of Energy (last visited Aug. 28, 2023).

[2] *Id.*

[3] *See* Danielle Vermeer, *Redlining and Environmental Racism*, Univ. of Mich. Sch. for Env't & Sustainability (Aug. 16, 2021), https://seas.umich.edu/news/redlining-and-environmental-racism; Jeremy Hoffman et al., *The Effects of Historical Housing Policies on Resident Exposure to Intra-Urban Heat: A Study of 108 US Urban Areas*, 8 Climate 12 (2020), attached as Exhibit 1.

[4] *Heat Vulnerability Index*, N.Y. State Dep't of Health (July. 2023), https://www.health.ny.gov/environmental/weather/vulnerability_index/.

Formerly redlined neighborhoods are as much as 7 degrees Celsius hotter than non-redlined neighborhoods, and this disparity is attributable to a lack of investments in built environment characteristics that reduce heat such as parks and green space.[5] People of color, who predominately live in formerly redlined neighborhoods such as Harlem and Bedford-Stuyvesant, have higher heat vulnerability not only because of neighborhood characteristics, but because people of color and low-income families are less able to access adequate cooling due to high energy burden.[6] Health impacts of extreme heat range from heat stress to heat stroke and even death.[7]

Q. What is the history of the environmental justice movement?

A. The concept of environmental justice was forged by a movement of scholars and activists responding to racial and class-based inequities in environmental policies and decision-making. The founders of the movement recognized that unfair and unequal treatment of communities of color and low-income people over the history of the United States led to disparities in exposure to environmental hazards.[8] Disparities in exposure to hazards means there are disparities in the impact on people's health and wellbeing.

Dr. Robert Bullard, known as the father of environmental justice, published the seminal report "Toxic Waste and Race" in 1987, which revealed that Black and Hispanic

[5] Hoffman et al. at 1–2, Ex. 1.

[6] *Mapping Inequality: Redlining in New Deal America*, Univ. of Richmond, https://dsl.richmond.edu/panorama/redlining/#loc=12/40.654/-74.064&city=brooklyn-ny (last visited Aug. 28, 2023); *see also* Sonal Jessel et al., *Energy, Poverty, and Health in Climate Change: A Comprehensive Review of an Emerging Literature*, 7 Frontiers 357 (2019) attached as Exhibit 2.

[7] Am. Pub. Health Ass'n & Ctrs. for Disease Control & Prevention, *Extreme Heat Can Impact Our Health in Many Ways* 1 (2016), https://www.cdc.gov/climateandhealth/pubs/extreme-heat-final_508.pdf.

[8] *See* Nat'l People of Color Env't Leadership Summit, *The Principles of Environmental Justice (EJ)* (1991), https://www.ejnet.org/ej/principles.pdf.

populations are exposed to more toxic contamination from hazardous waste sites than white neighborhoods.[9] In 1991 the People of Color Environmental Leadership Summit codified the Principles of the Environmental Justice Movement, stating that all public policy should be free from any form of discrimination or bias.[10] In 1994, the federal government recognized environmental justice concerns in Executive Order 12,898, which acknowledged the "disproportionately high and adverse human health or environmental effects of [the United States'] programs, policies, and activities on minority populations and low-income populations."[11]

III. ENVIRONMENTAL JUSTICE GUARANTEES IN THE CLCPA

Q. Are you familiar with the CLCPA?

A. Yes.

Q. Please summarize the greenhouse gas reduction mandates required in the CLCPA.

A. The CLCPA requires economy-wide greenhouse gas ("GHG") emissions reductions of 40% from 1990 levels by 2030, and 85% reductions from 1990 levels by 2050.[12] The law requires a zero-emission electric grid by 2040.[13] The law also requires 70% renewable electricity by 2030 and statewide energy efficiency gains of 185 trillion British thermal units ("tBtu") from the 2025 forecast.[14] The law covers all GHGs, including "carbon

[9] *See* Robert Bullard et al., *Toxic Wastes and Race at Twenty: Why Race Still Matters After All of These Years*, 38 Env't L. 371 (2008), https://heinonline.org/HOL/LandingPage?handle=hein.journals/envlnw38&div=19&id=&page=.

[10] Nat'l People of Color Env't Leadership Summit, *The Principles of Environmental Justice (EJ)*.

[11] *Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations*, 59 Fed. Reg. 7,629 (Feb. 11, 1994).

[12] ECL § 75-0107(1).

[13] PSL § 66-P(2).

[14] ECL § 75-0103(13)(e).

dioxide, methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons, sulfur hexafluoride, and any other substance emitted into the air that may be reasonably anticipated to cause or contribute to anthropogenic climate change."[15] The emissions reduction requirements cover not just emissions coming from electricity generation, but also combustion heating and transportation. The CLCPA does not specify GHG reduction sub-targets for different sectors.

Q. Please summarize how the CLCPA addresses other pollutants.

A. The CLCPA defines "co-pollutants" as "hazardous air pollutants produced by greenhouse gas emissions sources."[16] The CLCPA seeks to "maximize reductions of both greenhouse gas emissions and co-pollutants in disadvantaged communities . . . ."[17]

Q. Does the CLCPA address disadvantaged communities?

A. Yes.

Q. Please explain.

A. Section 7(3) of the CLCPA stipulates that state agencies, when making administrative approvals and decisions, cannot disproportionately burden DACs. Agencies must also prioritize reductions of GHGs and co-pollutants in disadvantaged communities.[18]

Q. Does the CLCPA provide any other requirements with respect to disadvantaged communities?

A. Section 75-0117 directs investments of funds. It stipulates that state agencies, authorities, and entities must:

[15] *Id*. § 75-0101(7).
[16] *Id*. § 75-0101(3).
[17] *Id*. § 75-0103(14)(d).
[18] *See* CLCPA § 7(3), S.B. 6599, 242d Sess. (N.Y. 2019).

> invest or direct available and relevant programmatic resources in a manner designed to achieve a goal for disadvantaged communities to receive forty percent of overall benefits of spending on clean energy and energy efficiency programs, projects or investments in the areas of housing, workforce development, pollution reduction, low income energy assistance, energy, transportation and economic development, provided however, that disadvantaged communities shall receive no less than thirty-five percent of the overall benefits of spending on clean energy and energy efficiency programs, projects or investments. . . .[19]

Q. How does the CLCPA ensure that disadvantaged communities are included in New York State's transition away from high emitting fossil fuels towards clean energy?

A. The CLCPA requires that DACs benefit from and are not disproportionately burdened by interventions to reduce emissions. The CLCPA also requires an equitable framework for addressing emissions reductions in disadvantaged communities. Section 7(3) is important because it will discontinue the pattern that disadvantaged communities have historically faced and currently face regarding higher burdens from climate change and other impacts of pollution from fossil fuels, including infrastructure, extraction, transmission, and distribution. Research shows that people of color and/or low-income individuals are more at-risk to climate change impacts and are exposed to more air pollution than the general population.[20] The New York City Panel on Climate Change's 2019 Report found that there is disproportionately more risk to climate change for low-income communities and

[19] ECL § 75-0117.

[20] *See* Diana Hernández, *Energy Insecurity: A Framework for Understanding Energy, the Built Environment, and Health Among Vulnerable Populations in the Context of Climate Change*, 103 Am. J. Pub. Health e32 (2013), attached as Exhibit 3; Christopher Tessum et al., *PM2.5 Polluters Disproportionately and Systemically Affect People of Color in the United States*, 7 Sci. Advances eabf4491 (2021), attached as Exhibit 4; Abdulrahman Jbaily et al., *Air Pollution Exposure Disparities Across US Population and Income Groups*, 601 Nature 228 (2022), attached as Exhibit 5.

households due to social, economic, and distributional inequity in environmental burdens and benefits.[21]

The CLCPA gives the Companies the obligation and opportunity to use the DAC maps (explained below) to provide an equal standard of service and investments for safe and reliable service, avoid increasing pollution or negative health outcomes in DACs, and target bill discounts and energy efficiency to customers in need. Low-income New Yorkers tend to pay a higher percentage of their incomes for energy ("energy burden") on average.[22] Additionally, research has documented low-income, Black, Latino/a, and Native American households face the highest energy burden in New York City.[23]

Q. You discussed disadvantaged communities. Can you please explain what a disadvantaged community is?

A. A "disadvantaged community" is a term used in the CLCPA to identify vulnerable communities in need of additional legal protections and investments during the coming decades as New York pursues its GHG emissions reduction goals. The CLCPA's umbrella definition of DAC is "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households . . . ."[24] However, the CLCPA also directed the CJWG to create a more

[21] Sheila Foster et al., *New York City Panel on Climate Change 2019 Report Chapter 6: Community-Based Assessments of Adaptation and Equity*, 1439 Annals N.Y. Acad. Scis. 126 (2019), attached as Exhibit 6.

[22] Am. Council for an Energy-Efficient Econ. ("ACEEE"), *Energy Burdens in New York City* (2020), https://www.aceee.org/sites/default/files/pdfs/aceee-01_energy_burden_-_new_york_city.pdf.

[23] *Id.*

[24] ECL § 75-0101(5).

detailed set of criteria based on this umbrella definition in order to identify the exact communities that qualify as a DAC across the tate.[25]

The CJWG issued draft DAC criteria in December 2021 and held a public comment period from March 9 to August 5, 2022.[26] The CJWG issued the final DAC criteria on March 27, 2023, after more than two years of deliberation.[27] New York State has also developed a map of DACs based on the final criteria.[28]

The final list of DACs encompasses communities in census tracts that scored in the top 28.9% of their regional or statewide percentile rank based on a combination of forty-five health, climate, environmental, and demographic geographic indicators.[29] Additionally, for the purposes of meeting the 35% investment floor, DACs include any households reporting annual total income at or below 60% of State Median Income or that are otherwise categorically eligible for low-income programs.[30]

Q. Can you explain what the Climate Justice Working Group is in relation to the CLCPA?

A. The CJWG was established according to section 75-0111 of the CLCPA. It is comprised of representatives from state agencies and environmental justice communities and tasked with establishing the criteria for DACs. The CLCPA defines Environmental Justice Community Representatives in this context as "members of communities of color, low-

[25] *Id*. § 75-0111(1)(b).
[26] *Disadvantaged Communities Criteria*, N.Y. State (2023), https://climate.ny.gov/Resources/Disadvantaged-Communities-Criteria.
[27] *Id*.
[28] *Id*.
[29] N.Y. State Dep't of Env't Conservation ("DEC"), *Climate Justice Working Group Meeting* at slide 28 (March 23, 2023), https://climate.ny.gov/-/media/Project/Climate/Files/CJWGMeeting03272023Presentation.pdf.
[30] *Id*. at slide 13.

income communities, and communities bearing disproportionate pollution and climate change burdens, or […] representatives of community-based organizations with experience and a history of advocacy on environmental justice issues . . . ."[31]

As required by the CLCPA, representatives from the following New York State agencies are present on the CJWG: the Department of Environmental Conservation ("DEC"), the Department of Health ("DOH"), the New York State Energy Research and Development Authority ("NYSERDA"), and the Department of Labor ("DOL"). As the law requires, the environmental justice community representatives include at least three representatives from New York City communities, three from rural communities, and three from upstate urban communities.[32] They are appointed by the State. Members of the working group are as follows:[33]

- Alanah Keddell-Tuckey, Chair, Director of the Office of Environmental Justice, DEC
- Neil Muscatiello, Director, Bureau of Environmental and Occupational Epidemiology, Center for Environmental Health, DOH
- Elizabeth Furth, Empire Fellow, DOL
- Christopher Coll, Director of Energy Affordability and Equity Program, NYSERDA
- Eddie Bautista, Executive Director, NYC Environmental Justice Alliance
- Jill Henck, Clean Energy Program Director, Adirondack North Country Association
- Sonal Jessel, Director of Policy, WE ACT for Environmental Justice
- Rahwa Ghirmatzion, Executive Director, PUSH Buffalo
- Amy Klein, CEO, Capital Roots
- Mary Beth McEwen, Interim Executive Director, Cornell Cooperative Extension of Oneida and Madison Counties
- Abigail McHugh-Grifa, Executive Director, Climate Solutions Accelerator of the Genesee-Finger Lakes Region

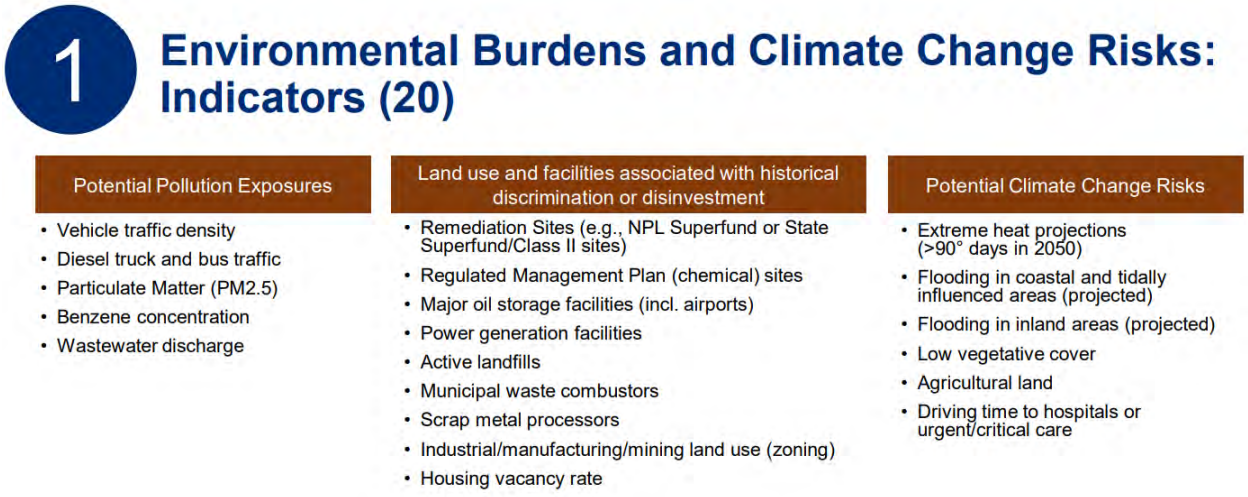
[31] ECL § 75-0111(1)(a).
[32] *Id*.
[33] *Climate Justice Working Group*, N.Y. State (2023), https://climate.ny.gov/Resources/Climate-Justice-Working-Group.

- Elizabeth Yeampierre, Executive Director, UPROSE

Q. What are the criteria for being considered a Disadvantaged Community?

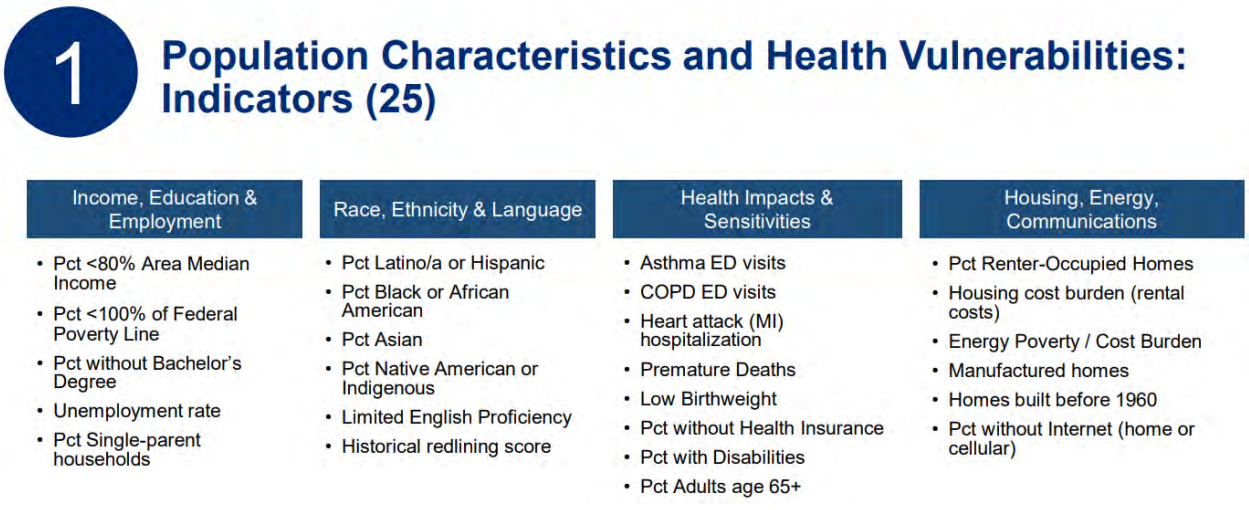
A. The CJWG used two sets of indicators to determine DACs: 1) environmental burdens and climate change risks; and 2) population characteristics and health vulnerabilities. The indicators are as follows:

Table 1. Environmental Burdens and Climate Change Risk



Reproduced from: DEC, *Climate Justice Working Group Meeting* at slide 20.

Table 2. Population Characteristics and Health Vulnerabilities



Reproduced from: DEC, *Climate Justice Working Group Meeting* at slide 21.

The indicators in Table 1 and Table 2 were geographically mapped and the census tracts at the top 28.9th percentile of the cumulative burden of these indicators were put forward by the working group as DACs for public consideration.[34] The DAC definition was limited to the top 28.9th percentile in order to achieve a maximum of 35% of the population being identified as a DAC.[35]

Q. Can you explain the map of disadvantaged communities?

A. The map is the result of selecting geographical criteria for defining DACs in New York State. It is a map that shows which census block groups are considered DACs. The intention of the map is to guide state decision making as required by the CLCPA.

Q: What are the goals of the CLCPA for environmental justice and disadvantaged communities?

A. In addition to the mandates discussed above, in consultation with the CJWG, New York State was required to prepare a program demonstrating community air monitoring systems in no less than four highest priority locations by October 1, 2022.[36] These are locations with high exposure burden, toxic contaminants, and criteria air pollutants. The air monitoring program has gone beyond four communities and is instead capturing information about air pollution and GHGs in the 10 highest polluting communities.[37] The

[34] DEC, *Climate Justice Working Group Meeting* at slide 20.
[35] *Id*. at slide 28.
[36] ECL § 75-0115(2)(b).
[37] DEC, *Climate Justice Working Group Meeting* at slide 13 (April 20, 2022), https://climate.ny.gov/-/media/Project/Climate/Files/2022-04-20-CJWG-Meeting-Presentation.pdf.

10 communities were selected by using a pollution burden score that was formulated by DEC for this purpose.[38] Additionally, the State must:

- Design and implement all regulations in a manner that seeks to be equitable, to minimize costs and to maximize the total benefits to New York, and encourages early action to reduce greenhouse gas emissions;
- Ensure that greenhouse gas emissions reductions achieved are real, permanent, quantifiable, verifiable, and enforceable by the department;
- Ensure that activities undertaken to comply with the regulations do not result in a net increase in co-pollutant emissions or otherwise disproportionately burden disadvantaged communities;
- Prioritize measures to maximize net reductions of [GHG] emissions and co-pollutants in disadvantaged communities . . . and encourage early action to reduce [GHG] emissions and co-pollutants; and
- Incorporate measures to minimize leakage.[39]

Q. Is there anything else?

A. Section 66-p states:

> In the allocation of ratepayer funds for clean energy, [the commission shall] direct the New York state energy research and development authority and investor owned utilities to develop and report metrics for energy savings and clean energy market penetration in the low and moderate income market and in disadvantaged communities, as defined in article seventy-five of the environmental conservation law, and post such information on the authority's website.[40]

IV. NEGATIVE HEALTH, ENVIRONMENTAL, SAFETY, AND ECONOMIC IMPACTS FROM THE COMBUSTION OF FOSSIL AND ALTERNATIVE FUELS

A. IMPACTS OF FOSSIL FUELS

Q. What are the negative health risks and outcomes associated with end-use appliances powered by the Companies' gas investments?

[38] *See id.*
[39] ECL § 75-0109(3).
[40] PSL § 66-p(7)(c).

A. As I will describe below, the combustion of natural gas used in buildings creates indoor and outdoor air pollution which negatively impacts health.

Q. Please describe the air pollutants released by gas appliances.

A. Gas stoves emit formaldehyde, carbon monoxide (CO), and nitrous oxides (NOx).[41] Within a few minutes of use, combustion from gas stoves without hoods create elevated levels of nitrogen dioxide (NO_2), beyond the U.S. Environmental Protection Agency outdoor safe level (100 ppb 1-hr average) and beyond the World Health Organization's recommended safe indoor level (106 ppb 1-hr average).[42]

Additionally, the emissions from natural gas-fired boilers and furnaces include NOx, CO, carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), volatile organic compounds (VOCs), trace amounts of sulfur dioxide (SO2), and particulate matter (PM or PM2.5).[43]

Q. Please describe how this air pollution impacts health.

A. NOx is nitrogen dioxide (NO_2) or nitrogen oxide (NO) and is the principal poison created by gas stove combustion.[44]

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[41] *Indoor Air Pollution From Cooking*, Cal. Air Res. Bd., https://ww2.arb.ca.gov/resources/documents/indoor-air-pollution-cooking#:~:text=Natural%20gas%20and%20propane%20stoves,air%20pollution%20from%20wood%20smoke (last visited Aug. 28, 2023); Brady Seals & Andee Krasner, Rocky Mountain Inst. et al., *Health Effects from Gas Stove Pollution* 7 (2020), attached as Exhibit 7; WE ACT for Environmental Justice et al., *Out of Gas, in with Justice* 15–16 (2023), attached as Exhibit 8.

[42] Eric Lebel et al., *Methane and NOx Emissions from Natural Gas Stoves, Cooktops, and Ovens in Residential Homes*, 56 Env't Sci. Tech. 2529, 2529 (2022), attached as Exhibit 9; Seals & Krasner at 11, Ex. 7.

[43] U.S. Env't Prot. Agency ("EPA"), *1.4: Natural Gas Combustion* 2 (2020), https://www.epa.gov/sites/default/files/2020-09/documents/1.4_natural_gas_combustion.pdf.

[44] Seals & Krasner at 8, Ex. 7.

The U.S. Environmental Protection Agency has linked NOx exposure to asthma.[45] Asthma can lead to lifelong impairments, hospitalization, or death. Other health effects of NOx exposure include: lower IQ in children, increased inflammation of the airways, worsened cough and wheezing, reduced lung function, increased asthma attacks, greater likelihood of emergency department and hospital admissions, low birthweight, premature death, and cardiovascular illness.[46]

Indoor fuel combustion in New York poses grave health threats. In 2017, 1,940 premature deaths were caused by fuel combustion in buildings in New York, more than in any other state.[47] The vast majority of those deaths, 1,114, took place in New York City.[48]

Black and Latino/a children, who have higher rates of childhood asthma than children of other races in New York City, are likely the most disproportionately burdened by indoor air pollution from gas stoves.[49] Asthma in New York City is highest in communities of color, such as East Harlem and the South Bronx.[50] Ashma disproportionately impacts Black children: one study found that a non-Hispanic Black

[45] EPA, *Environments and Contaminants: Criteria Air Pollutants* 3–4 (2015), https://www.epa.gov/sites/default/files/2015-10/documents/ace3_criteria_air_pollutants.pdf.

[46] Seals & Krasner at 13, Ex. 7; *Nitrogen Dioxide*, Am. Lung Ass'n, https://www.lung.org/clean-air/outdoors/what-makes-air-unhealthy/nitrogen-dioxide (last updated Apr. 17, 2023).

[47] Talor Gruenwald & Stephen Mushegan, *New York Emits More Building Air Pollution Than Any Other State*, Rocky Mountain Inst. (May 18, 2021), https://rmi.org/new-york-emits-more-building-air-pollution-than-any-other-state/.

[48] *Id*.

[49] NYC Health, *Disparities Among Children with Asthma in New York City* 1 (2021), https://www1.nyc.gov/assets/doh/downloads/pdf/epi/databrief126.pdf; Seals & Krasner at 13, Ex. 7.

[50] NYC Health, *Disparities Among Children with Asthma in New York City* at 2.

child is nearly eight times more likely to die from an asthma attack than a non-Hispanic white child.[51]

The NOx emissions from gas stoves are more dangerous in homes that do not have an adequately functioning ventilation hood or have poor ventilation.[52] Black, Latino/a, and Asian/Pacific Islander households reported mold in their homes at higher rates than white households, which points to a potential disparity in adequately ventilated homes.[53]

Formaldehyde is a known human carcinogen and simmering a gas stove at low levels has been found to produce significant exposure.[54] CO is a gas that is emitted from gas stoves and can cause premature mortality.[55]

PM2.5 is emitted through gas stove flames or cooking food and is a leading cause of respiratory and cardiovascular illness.[56] Particulate matter is the largest environmental health risk in the United States and is associated with increased mortality rates from cardiovascular disease, chronic obstructive pulmonary disease, and lung cancer.[57]

Q. Are there particular risks for disadvantaged communities?

[51] WE ACT at 16, Ex. 8.
[52] Lebel et al. at 2529–30, Ex. 9.
[53] *Living and Environmental Conditions, Subtopic Housing: Mold*, NYC Health, https://a816-health.nyc.gov/hdi/epiquery/visualizations?PageType=tsi&PopulationSource=CHS&Topic=7&Subtopic=14&Indicator=Mold&Year=2012 (last visited Aug. 28, 2022).
[54] Seals & Krasner at 8, Ex. 7.
[55] *Id*. at 8–10, Ex. 7.
[56] Seals & Krasner at 8, Ex. 7; EPA, *Environments and Contaminants: Criteria Air Pollutants* at 2–3.
[57] Maninder P.S. Thind et al., *Fine Particulate Air Pollution from Electricity Generation in the US: Health Impacts by Race, Income, and Geography*, 53 Env't Sci. Tech. 14,010 (2019), attached as Exhibit 10.

A. Yes. The status quo is a result of environmental racism and it disproportionately burdens DACs.

By definition, DACs are neighborhoods experiencing high pollution burdens, as that is how these areas became designated as DACs.[58] The New York City Department of Health and Mental Hygiene has found that air pollution particularly affects residents of low-income neighborhoods: one study concluded that the rate of emergency room visits among children due to PM2.5 attributable asthma is three times higher in high poverty compared to low poverty neighborhoods.[59]

Environmental pollution and negative health burdens also fall along disparate racial lines. A 2019 analysis found that Asian, Latino/a, and Black New Yorkers are exposed to higher levels of particulate matter from cars, trucks, and buses than are white New Yorkers.[60] 74% of New York's Black population and nearly 80% of the state's Asian population live in areas where on-road particulate matter concentrations attributable to transportation exceed state averages. Asian New Yorkers are exposed to twice as much particulate matter pollution from vehicles compared to white New Yorkers.[61] Latino/a and Black New Yorkers are exposed to 81% and 72%, respectively, more particulate matter pollution from vehicles compared to white New Yorkers.[62]

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[58] DEC, *Climate Justice Working Group Meeting* at slide 20.

[59] Iyad Kheirbek et al., N.Y.C. Dep't of Health & Mental Hygiene, *Air Pollution and the Health of New Yorkers: The Impact of Fine Particles and Ozone* 23 (2011), https://www1.nyc.gov/assets/doh/downloads/pdf/eode/eode-air-quality-impact.pdf.

[60] Union of Concerned Scientists, *Inequitable Exposure to Air Pollution from Vehicles in New York State* 1 (2019), https://www.ucsusa.org/sites/default/files/attach/2019/06/Inequitable-Exposure-to-Vehicle-Pollution-NY.pdf.

[61] *Id*.

[62] *Id*.

Emissions of nitrogen dioxide from diesel traffic—a precursor to ozone, which can cause cardiovascular and respiratory illness—also disproportionately harm people who are Black, Hispanic or Latino/a, Asian, and Native American in New York City.[63] This pollution contributes to the cumulative impacts of the gas distribution system.

DACs are experiencing higher rates of air pollution and therefore chronic illness. Continued exposure to pollution from gas stoves, other gas appliances, and the Companies' and others' polluting facilities in neighborhoods puts people living in DACs at greater risk of negative health and economic outcomes. For this reason, the CLCPA requires state agencies to specifically focus on reducing pollution in DACs.

Q. Are DACs disproportionately burdened in any other way?

A. Yes. DACs are more likely to experience energy insecurity, which brings its own health impacts.

Q. Please describe how energy insecurity impacts health.

A. Energy insecurity results in many negative health outcomes. Over 25% of New York City households have more than a 6% energy burden.[64] Black, then Latino/a households reported facing the most energy insecurity in the United States.[65] 35% of non-Hispanic

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[63] Mary Angelique G. Demetillo et al., *Space-Based Observational Constraints on NO_2 Air Pollution Inequality from Diesel Traffic in Major US Cities*, 48 Geophysical Rsch. Letters 3–4 (2021), attached as Exhibit 11; *see also* N.Y. State Dep't of Health, *New York's State Health Improvement Plan: Prevention Agenda 2019-2024* at 72–73 (last updated June 30, 2023), https://www.health.ny.gov/prevention/prevention_agenda/2019-2024/docs/ship/nys_pa.pdf (noting the "[e]xtensive evidence" linking ozone with respiratory and cardiovascular illness and death and establishes a goal to "[r]educe exposure to outdoor air pollutants," with an emphasis on vulnerable groups).

[64] ACEEE, *Energy Burdens in New York City* (2020) ("A high energy burden is considered to be above 6%, and a severe energy burden above 10%").

[65] Ariel Drehobl & Lauren Ross, ACEEE, *Lifting the High Energy Burden in America's Largest Cities: How Energy Efficiency Can Improve Low Income and Underserved Communities*

Black households reported facing energy insecurity compared to 14% of non-Hispanic white households.[66]

Amongst Black and low-income New Yorkers, the number one barrier to accessing adequate cooling to stay safe in heatwaves is affording the utility bill costs of running an air conditioner.[67] Modeling done by researchers at Columbia University predicts that by 2080 there will be over 3,000 heat-related deaths per year if conditions do not change,[68] and likely the disparity in deaths by race will continue without targeted intervention. Black New Yorkers are twice as likely to die from extreme heat due to a lack of access to adequate energy for cooling, which is a result of structural racism.[69]

Sometimes when people cannot access adequate energy, they turn to maladaptive coping mechanisms.[70] For example, a study found that low-income mothers turn on their gas stoves to heat their home when their utility provider or building owner has not provided adequate heating.[71] This coping mechanism exposes the mother and children to NOx levels between 130–546 ppb at its peak, which far exceeds healthy levels.[72]

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4 (2016), https://www.aceee.org/sites/default/files/publications/researchreports/u1602.pdf.

[66] Diana Hernández et al., Columbia Univ., Nat'l Ctr. for Children in Poverty, *Energy Insecurity Among Families with Children* 4 (2014), https://www.nccp.org/wp-content/uploads/2020/05/text_1086.pdf.

[67] NYC Health, *2023 NYC Heat-Related Mortality Report* (2023), https://a816-dohbesp.nyc.gov/IndicatorPublic/beta/key-topics/climatehealth/heat-report/.

[68] Elisaveta Petkova et al., *Towards More Comprehensive Projections of Urban Heat-Related Mortality: Estimates for New York City under Multiple Population, Adaptation, and Climate Scenarios*, 125 Env't Health Persp. 47–55 (2017), attached as Exhibit 12.

[69] NYC Health, *2023 NYC Heat-Related Mortality Report*.

[70] *See* Hernández, *Energy Insecurity Among Families with Children* at 7.

[71] *Id.*

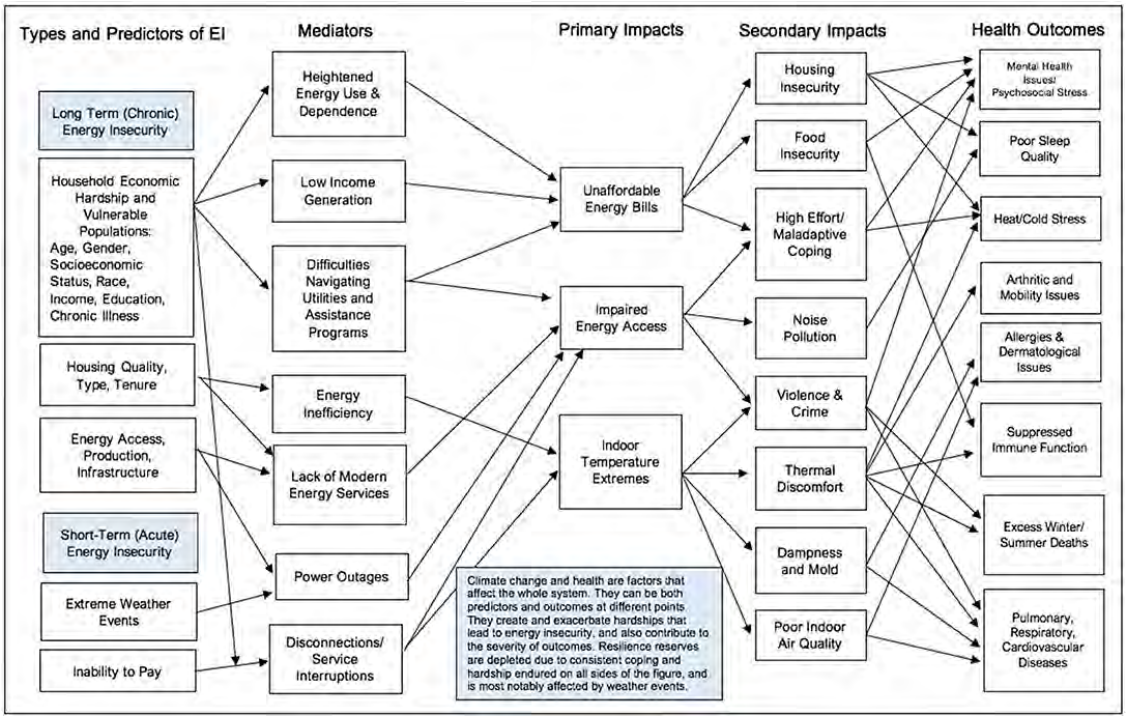
[72] Seals & Krasner at 11, Ex. 7; *see also* Martine Vrijheid et al., *Prenatal Exposure to Gas Cooking and Neurodevelopment at 14 Months*, 20 Epidemiology S37 (2009), attached as Exhibit 13.

Additionally, energy insecurity from power outages, planned outages, shut-offs, impaired access, or inability to pay has health impacts.[73] These outages can cut off access to necessary medical devices, spoil healthy and expensive foods, and lead to hyper- or hypothermia.[74] Energy insecurity has also been linked to suppressed immune function, carbon monoxide poisoning from maladaptive coping mechanisms, arthritic and mobility impairment, poor sleep, allergies, respiratory and cardiovascular disease, and excess summer and winter deaths.[75] Households that have experienced shutoffs report more mental health issues and psychosocial stress.[76]

Energy insecurity has many other both direct and indirect health outcomes as exemplified in the figure below.

[73] *See* Jessel et al. at 5–6, Ex. 2.
[74] *Id.*
[75] *Id.* at 9, 11–12.
[76] *Id.* at 12.

Figure 1: Connection Between Household Energy Insecurity (EI) and Health



Reproduced from: Jessel et al. at 4 fig.2.

B. IMPACTS OF RENEWABLE NATURAL GAS

Q. Are you familiar with Renewable Natural Gas?

A. Yes

Q. What is Renewable Natural Gas?

A. Renewable Natural Gas ("RNG") is another term for biomethane, which is treated biogas that can be used as a substitute for fossil gas.[77] Anaerobic digestion, the decomposition of wet, organic matter by microorganisms in an oxygen-free environment, is often used to produce biogas from organic sources including crops, food waste, and animal manure.[78]

[77] *Renewable Natural Gas*, EPA, https://www.epa.gov/lmop/renewable-natural-gas (last updated Aug. 3, 2023).

[78] Sasan Saadat et al., *Rhetoric vs. Reality: The Myth of "Renewable Natural Gas" for Building Decarbonization* 6 (2020) ("Myth of RNG Report"), https://earthjustice.org/wp-content/uploads/report_building-decarbonization-2020.pdf.

Q. Are the Companies proposing ratepayer-funded investments in fuels other than fossil fuels?

A. Yes. Both Companies are proposing ratepayer-funded investments in infrastructure for RNG. KEDNY has requested recovery of $14.003 million in capital investments for two RNG interconnection projects as well as $1.000 million in Data Year 3 for future RNG interconnections.[79] KEDNY has also requested recovery of a $15.713 million investment at the Newtown Creek RNG facility.[80] KEDLI has requested recovery of $11.209 million in capital investments for two RNG interconnection projects as well as $1.000 million in Data Year 3 for future RNG interconnections.[81]

Q. Are there any health impacts with replacing fossil natural gas with RNG in end uses such as stoves and ovens?

A. Yes.

Q. Can you please explain the health impacts of using RNG with end-use appliances?

A. Because RNG is primarily methane, its combustion will produce the same local pollutants as fossil natural gas, including, as discussed above in Section IV(A), NOx, PM2.5 and other pollutants that cause serious health harms.[82]

Additionally, because RNG feedstocks such as food waste contain high levels of persistent contaminants such as per- and polyfluoroalkyl substances ("PFAS"),[83] if RNG

[79] Case No. 22-G-0225, Gas Infrastructure and Operations Panel at 78, Brooklyn Union Gas Co., N.Y. Dep't of Pub. Serv. (Apr. 28, 2023) (Dkt. No. 2) ("KEDNY GIOP").
[80] *Id*. at Exhibit __ (GIOP-1) at 1.
[81] Case No. 22-G-0226, Gas Infrastructure and Operations Panel at 74, KeySpan Gas East Corp., N.Y. Dep't of Pub. Serv. (Apr. 28, 2023) (Dkt. No. 2) ("KEDLI GIOP").
[82] Myth of RNG Report at 16.
[83] *See, e.g.*, Off. of Rsch. & Dev., EPA, *Emerging Issues in Food Waste Management* (2021), https://www.epa.gov/system/files/documents/2021-08/emerging-issues-in-food-waste-management-persistent-chemical-contaminants.pdf.

conditioning does not remove PFAS, then RNG use in homes could expose customers to PFAS air emissions. Exposure to PFAS can result in developmental toxicity, immunotoxicity, hormonal disruption, and cancer.[84] PFAS are emerging contaminants for which the regulatory structure is currently in development, but they are dangerous even in very low levels.[85] The Companies' gas quality requirements do not include standards for PFAS,[86] and they have not demonstrated that the RNG they plan to purchase at the proposed interconnection sites will be conditioned in a way that will remove all PFAS.

Q. Does RNG production have any health or environmental impacts?

A. Facilities where fossil gas alternatives such as RNG are produced can exacerbate air and water pollution impacts in nearby communities.[87] Anaerobic digestion has been identified as a significant source of ammonia emissions, which can contribute to the formation of particulate matter.[88] As discussed above, particulate matter is a dangerous pollutant that causes serious health impacts.

The developers from which the Companies plan to purchase RNG each need to obtain a Title V permit, indicating that the planned anaerobic digestion and biogas conditioning will produce air emissions.[89] Additionally, creating a market for RNG can

[84] *Id*. at 4.
[85] For example, the U.S. Environmental Protection Agency has set advisory levels for PFAS in water at the extremely low level of 70 parts per trillion. *See* Kevin Loria, *EPA Says Even Extremely Low Levels of PFAS in Drinking Water May Be Unsafe*, CR (June 15, 2022).
[86] *See* Companies' Response to WE ACT Interrogatory 71 Attachment 1, attached as Exhibit 14; Companies' Response to WE ACT Interrogatory 72 Attachment 1, attached as Exhibit 15.
[87] Myth of RNG Report at 2.
[88] Michael W. Bell et al., *Ammonia Emissions from an Anaerobic Digestion Plant Estimated Using Atmospheric Measurements and Dispersion Modelling* 56 Waste Mgmt. 113 (2016), attached as Exhibit 16.
[89] *See* Companies' Response to DPS Interrogatory 354, attached as Exhibit 17.

incentivize activities that create methane, resulting in a net increase in methane emissions.[90]

Q. Does replacing fossil gas in buildings with RNG reduce greenhouse gas emissions?

A. No. In fact, replacing fossil gas in buildings with RNG will only postpone the transition to a truly healthy and sustainable building stock that relies on electrification.

Like fossil gas, RNG is primarily methane, so its leakage and combustion emit the same level of GHGs as fossil gas.[91] RNG created from intentionally produced methane always results in net GHG emissions increases because of methane leakage during transport.[92] Even if the Companies invest in repairing or replacing leak prone pipes, some amount of leakage is inevitable.[93]

Additionally, NYSERDA has confirmed that combustion emissions from RNG must be accounted for when comparing emissions reductions with the CLCPA reduction mandates.[94] It is therefore inappropriate for the Companies to claim that the proposed RNG interconnections will eliminate GHG emissions from displaced fossil gas that would have been combusted downstream.[95] The Companies and the Commission must account for the emissions that take place at the RNG production site that are an inevitable

[90] Myth of RNG Report at 9.
[91] *Id*. at 16.
[92] Emily Grubert, *At Scale, Renewable Natural Gas Systems Could be Climate Intensive: The Influence of Methane Feedstock and Leakage Rates* 15 Env't Rsch. Letters (2020), attached as Exhibit 18.
[93] *Id*.
[94] N.Y. State Energy Rsch. & Dev. Auth. ("NYSERDA"), *Fossil and Biogenic Fuel Greenhouse Gas Emission Factors* 2, 5 (2022), https://www.nyserda.ny.gov/-/media/Project/Nyserda/Files/Publications/Energy-Analysis/22-23-Fossil-and-Biogenic-Fuel-Greenhouse-Gas-Emission-Factors.pdf.
[95] KEDNY GIOP at 80; KEDLI GIOP at 77.

consequence of purchasing RNG for use in the gas distribution system, as well as combustion emissions.

The Companies' proposed use of RNG will also result in out-of-state leakage, and therefore not achieve GHG reductions. The CLCPA defines leakage as "a reduction in emissions of greenhouse gases within the state that is offset by an increase in emissions of greenhouse gases outside of the state."[96] The Companies have stated that they do not intend to purchase the environmental attributes associated with the four proposed new RNG interconnections,[97] and that KEDNY will sell the environmental attributes associated with the Newtown Creek RNG facility.[98] The environmental attributes for these projects will therefore be purchased by out-of-state fuel suppliers for compliance with a different regulatory system outside of New York. As a result, the purported GHG benefits associated with the RNG will be monetized in a different market, allowing fuel sellers outside of New York to avoid meaningful emissions reductions by purchasing environmental attributes made in New York. To the extent that replacing fossil gas with RNG produces any GHG benefits, therefore, those benefits will be erased by GHG emissions elsewhere.

Q. The Companies state that the proposed RNG interconnections "support the CLCPA goals of reducing GHG emissions from the heating sector."[99] Is this correct?

[96] *See* ECL § 75-0101(12) (defining "leakage"); *see also id*. §§ 75-0103(13)(k) (directing the Climate Action Council to devise mechanisms to minimize leakage), 75-0109(3)(e) (directing DEC to design regulations to avoid leakage).
[97] Companies' Response to DPS Interrogatory 455, attaches as Exhibit 19.
[98] KEDNY GIOP at 83; Companies' Response to DPS Interrogatory 780, attached as Exhibit 20.
[99] KEDNY GIOP at 81; KEDLI GIOP at 77.

A. No. As an initial matter, the CLCPA sets economywide emissions reduction mandates, not sector-by-sector goals.[100] In December 2022 the Climate Action Council issued the final Scoping Plan, which sets forth policy recommendations for each sector that must form the basis for regulations promulgated under the CLCPA and the State Energy Plan.[101] The Companies' proposed RNG interconnections are not consistent with the Scoping Plan.

The Scoping Plan specifies that before alternative fuels are considered for the gas system, it is important to ensure that "there are demonstrated . . . GHG benefits" and that "any use of alternative fuels, including green hydrogen and renewable natural gas, does not result in increased leaks and emissions within the entire life cycle of the development and use of alternative fuels."[102] The Companies have not demonstrated that the use of RNG will result in greenhouse gas benefits and will not result in increased leaks or lifecycle emissions.

Using RNG in buildings conflicts with the Scoping Plan by prolonging reliance on the gas distribution system. The Scoping Plan envisions that by 2030, 1–2 million energy efficient homes will be electrified with heat pumps and that heat pumps will provide space heating and cooling for 10–20% of commercial space statewide.[103]

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[100] ECL § 75-0107(1).

[101] *See id*. §§ 75-0103(11); 75-0109(2)(c); N.Y. State Climate Action Council, *Final Scoping Plan* (Dec. 2022), https://climate.ny.gov/Resources/Scoping-Plan (hereinafter "Scoping Plan").

[102] Scoping Plan at 361.

[103] *Id*. at 179; *see also* NYSERDA, *Governor Hochul Announces Plan to Achieve 2 Million Climate-Friendly Homes by 2030* (Jan. 5, 2022), https://www.nyserda.ny.gov/About/Newsroom/2022-Announcements/2022-01-05-Governor-Hochul-Announces-Plan-to-Achieve-2-Million-Climate-Friendly-Homes-By-2030 (stating Governor Hochul's plans to achieve two million electrified or electrification-ready homes by 2030).

Additionally, the Scoping Plan specifies that the majority of new purchases for space and water heating should be heat pumps starting this decade, and from 2030 on, more than 250,000 homes should be retrofitted or constructed to be energy efficient and ready to install heat pumps for heating, cooling, and hot water.[104] The Scoping Plan also determined that by 2050, 85% of homes and commercial spaces should be electrified with efficient heat pumps and thermal energy networks.[105]

The Scoping Plan sets out benchmarks that existing buildings must meet to achieve the CLCPA's requirements, which make clear that in the coming years the State must move away from reliance on gas. The Scoping Plan calls for phasing out fossil fuel combustion in existing buildings by prohibiting the replacement of fossil fuel combustion equipment—requiring the installation of non-fossil fuel alternatives instead—for heating, cooling, and hot water by 2030 (for small residential customers) and 2035 (for large residential and commercial customers).[106] The Scoping Plan further recommends prohibiting replacement of fossil fuel equipment for cooking and clothes drying by 2035.[107]

Q. Does the Scoping Plan say anything about the gas distribution system?

A. Yes. The Scoping Plan confirms the need to downsize the gas distribution system in the coming years.[108] The Climate Action Council recommends utility-specific forecasts for reducing gas sales and decreasing the number of gas customers over time as well as applying greater scrutiny to investments in the gas distribution system to ensure that

[104] Scoping Plan at 179.
[105] *Id*. at 180.
[106] *Id.* at 190.
[107] *Id*.
[108] *Id.* at 350.

investments do not result in stranded assets and make it even more expensive to reduce GHGs in the future.[109] As a result, under the CLCPA, New York's gas distribution system will need to downsize and gas usage will necessarily decrease. The Companies' assertion that RNG will reduce greenhouse gas emissions by replacing fossil gas that would otherwise be used, therefore, is wholly untethered from New York's policy reality.

Nor can the gas system be propped up by RNG. The Scoping Plan treats RNG with caution, for good reason.[110] RNG is far more expensive than fossil gas and is not widely available enough to replace fossil gas. A recent report by Groundwork Data found that replacing fossil gas with RNG across New York would cost ratepayers $25 billion annually on top of the current expense of maintaining the gas pipeline system.[111] Even in a hybrid scenario that relies on a mix of electrification and gas pipelines, high RNG costs will drive up customers' gas bills.[112] Moreover, in a hybrid scenario, per-customer costs will increase as fewer ratepayers shoulder the costs of maintaining the distribution system.[113]

Because exiting the gas system requires upfront investments in energy efficiency and electrification measures, the costs of a hybrid system would increasingly fall on low-income ratepayers.[114] By contrast, a managed transition that moves away from pipeline gas in phases through targeted pruning—and frontloads investments in low- to moderate-

[109] *Id.* at 360, 362.
[110] *See id.* at 351 (noting that the availability of RNG is currently small and urging analysis of full lifecycle GHG and co-pollutant emissions impacts).
[111] Groundwork Data & Building Decarbonization Coal., *The Future of Gas in New York State* 40 (2023) (hereinafter "BDC Report"), https://buildingdecarb.org/wp-content/uploads/BDC-The-Future-of-Gas-in-NYS.pdf.
[112] *Id.* at 46–48.
[113] *Id.* at 47.
[114] *Id.* at 26–27, 46–48.

income households and DACs—will avoid ballooning costs and minimize burdens to ratepayers.[115]

Groundwork Data's findings are supported by a report published this past June by the American Council for an Energy-Efficient Economy ("ACEEE"), which studied the impacts of electrification and gas decarbonization on distribution costs nationwide.[116] ACEEE concluded that an unmanaged transition in which electrification occurs on a household-by-household basis is the costliest scenario, whereas strategic pipeline replacement and electrification would reduce costs to ratepayers.[117] Replacing a gas distribution line and helping customers electrify could save utilities up to $1,300 per customer, according to ACEEE.[118] By contrast, investing in alternative fuels that will prolong the gas system will lead to increased utility bills for the customers who can least afford them, in contravention of the Commission's mandate to ensure just and reasonable rates.[119]

Q. Do you know the cost of the RNG that the Companies inject or propose to inject into the distribution system?

A. The price per dollar of million British thermal units ("MMBtu") of RNG that the Companies currently inject into the system is confidential,[120] and the Companies have not finalized negotiations with the developers of the four new proposed RNG interconnection

[115] *Id*. at 53–54.
[116] *See* Steven Nadel, *Impact of Electrification and Decarbonization on Gas Distribution Costs*, ACEEE (June, 2023), https://www.aceee.org/sites/default/files/pdfs/U2302.pdf.
[117] *Id*. at iv, 17, 27.
[118] *Id*. at 17.
[119] PSL § 65(1).
[120] Companies' Response to WE ACT Interrogatory 105(d), attached as Exhibit 21; Companies' Response to WE ACT Interrogatory 109, attached as Exhibit 22.

sites.[121] However, according to the Groundwork Data report, projected production costs for RNG produced in New York State range from $10 to more than $50 per million Btu, as compared to $3.50 to $5 per million Btu for fossil gas, because of limited feedstocks and the energy required to produce RNG.[122] A report for NYSERDA estimated average costs between $11.29 MMBtu and $34.56 MMBtu.[123] Even putting aside climate and public health concerns, cost barriers disqualify RNG as a viable decarbonization solution for the gas system.

For these reasons, using RNG to replace fossil gas in the existing pipeline system is a dead end. RNG should be conserved for sectors that cannot be electrified; the building sector is not one of them.

C. IMPACTS OF HYDROGEN

Q. Are you familiar with green hydrogen?

A. Yes.

Q. What is green hydrogen?

A. Green hydrogen is made through a process called electrolysis that uses renewable electricity to split hydrogen from water molecules.[124]

Q. Are the Companies proposing ratepayer-funded investments in green hydrogen?

[121] Companies' Response to WE ACT Interrogatory 108, attached as Exhibit 23.
[122] BDC Report at 28.
[123] ICF Res. LLC, *Potential of Renewable Natural Gas in New York State* at iii (2022), https://www.nyserda.ny.gov/-/media/Project/Nyserda/files/EDPPP/Energy-Prices/Energy-Statistics/RNGPotentialStudyforCAC10421.pdf.
[124] Sasan Saadat & Sara Gersen, *Reclaiming Hydrogen for a Renewable Future: Distinguishing Oil & Gas Industry Spin from Zero-Emission Solutions*, 3, 13 (Aug. 2021) https://earthjustice.org/wp-content/uploads/hydrogen_earthjustice_2021.pdf.

A. Yes. KEDLI is requesting $13.291 million in capital costs for the HyGrid Project, a pilot demonstration project to enable hydrogen blending as part of a green hydrogen facility in the Town of Hempstead, New York.[125] The project is planned to be operational in March 2025 and the hydrogen will be injected into the existing distribution system that provides gas service to 844 Hempstead customers.[126] The company plans to begin the blend rate at 3% by volume and increase it to 20% over an eighteen-month period.[127]

Q. What are the potential health impacts of hydrogen blending?

A. Hydrogen combustion produces up to six times as much NOx as methane combustion.[128] As discussed above, NOx can cause asthma and other harmful health conditions. As a result, combusting hydrogen in end-use appliances could pose heightened threats to public health.

Q. Does hydrogen blending reduce GHGs in the building sector?

A. Hydrogen blending cannot achieve meaningful GHG reductions in the building sector and is not a viable decarbonization pathway. Only very low levels of hydrogen can be blended into natural gas systems. In 2022, the California Public Utilities Commission ("CPUC") completed a hydrogen blending feasibility study that involved a literature

[125] KEDLI GIOP Testimony at 81.
[126] *Id*. at 81-82.
[127] *Id.* at 83; Companies' Response to NRDC Interrogatory 27(a), attached as Exhibit 24.
[128] Mehmet Salih Cellek & Ali Pinarbasi, *Investigations on Performance and Emission Characteristics of an Industrial Low Swirl Burner While Burning Natural Gas, Methane, Hydrogen-Enriched Natural Gas and Hydrogen as Fuels*, 43 Int'l J. of Hydrogen Energy 1194, 1205 (2018), https://www.sciencedirect.com/science/article/abs/pii/S0360319917319791; *see also* Rachel Parkes, *Physicians' Groups Warn Burn of H2 and Natural Gas Blends in Heating Raises Chances of Deadly Explosions, as well as Exacerbating Asthma and Dementia Hazards*, Recharge (June 23, 2022), https://www.rechargenews.com/energy-transition/hydrogen-blending-with-natural-gas-puts-lives-at-risk-us-doctors/2-1-1244492.

review, modeling, and experimental work.[129] The CPUC has concluded that "[t]he systemwide blending injection scenario becomes concerning as hydrogen blending approaches five perfect by volume" due to limitations with both the distribution system and end-use appliances.[130] A 5% hydrogen blend would correspond to an energy contribution of less than 2%.[131] The result is that hydrogen blending would have a minimal impact, if any, on overall GHG emissions.[132] Even at the higher blending levels proposed by the Companies, hydrogen would only reduce the carbon intensity of natural gas from 73.1 to 67.8 kilograms of carbon dioxide equivalent per million btu.[133]

Additionally, because hydrogen is a small molecule it is more likely to leak than methane.[134] Hydrogen leakage reduces any potential climate benefits because hydrogen acts as an indirect greenhouse gas in the atmosphere by increasing the amounts of other gases such as methane.[135]

Q. Do you have any other concerns about hydrogen blending?

A. Yes. As I mentioned, the CPUC concluded that hydrogen blends higher than 5% would raise safety concerns, and I am concerned that KEDLI's plan to ramp up to 20% is unsafe. Additionally, hydrogen is more likely to leak than methane. Hydrogen leakage

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[129] *See* Cal. Pub. Utilities Comm'n, R.13-02-008, *Hydrogen Blending Impacts Study* 4 (2022), (hereinafter "CPUC Report"), https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M493/K760/493760600.PDF.

[130] CPUC Report at 4.

[131] BDC Report at 29.

[132] *Id.*

[133] Companies' Response to NRDC Interrogatory 28(a), attached as Exhibit 25.

[134] CPUC Report at 3, 11.

[135] *See* Ilissa Ocko & Steven Hamburg, *Climate Consequences of Hydrogen Emissions* 22 Atmospheric Chemistry & Physics 9349 (2022), https://acp.copernicus.org/articles/22/9349/2022/acp-22-9349-2022.pdf; Maria Sand et al., *A Multi-model Assessment of the Global Warming Potential of Hydrogen* 4 Commc'ns Earth & Env't (2023), https://www.nature.com/articles/s43247-023-00857-8.

poses safety risks because hydrogen can easily ignite.[136] Allowing hydrogen blending is therefore inconsistent with the Commission's obligation to provide safe and adequate service.[137]

Moreover, like RNG, hydrogen combustion can only prolong the necessary downsizing of the gas distribution system and interfere with the managed transition that we need to avoid ballooning utility bills and stranded assets.

Q. Do you have additional concerns about the HyGrid project?

A. Yes. Although KEDLI presents HyGrid as a non-pipeline alternative, the Company has acknowledged that "HyGrid does not defer, avoid, or eliminate the need for any specific planned traditional infrastructure project on the Company's distribution system."[138] Additionally, the Company admits that the HyGrid project is not likely to meet the acceptance criteria for benefit-cost analysis.[139] KEDLI is apparently counting on hydrogen becoming cost-competitive through the U.S. Department of Energy's "Hydrogen Shot" initiative, an ambitious effort to reduce the cost of green hydrogen that the Department of Energy recognizes is based on "stretch" research and development goals.[140] In other words, green hydrogen is a nascent technology and its cost competitiveness is far from guaranteed—even more so given that the HyGrid project is not eligible for federal support from the Inflation Reduction Act.[141]

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[136] CPUC Report at 7.
[137] PSL § 65(1).
[138] Companies' Response to DPS Interrogatory 693, attached as Exhibit 26.
[139] *Id*.
[140] *Id*.; U.S. Dep't of Energy, *Pathways to Commercial Liftoff: Clean Hydrogen* 2 (2023), https://liftoff.energy.gov/wp-content/uploads/2023/05/20230523-Pathways-to-Commercial-Liftoff-Clean-Hydrogen.pdf.
[141] Companies' Response to WE ACT Interrogatory 99, attached as Exhibit 27.

At bottom, hydrogen is a tool to maintain the gas distribution system and will frustrate rather than facilitate the transition to a truly clean energy economy through proven cost-effective and health-protective measures such as electrification and energy efficiency.

V. THE COMPANIES MUST PRIORITIZE REDUCING GHG EMISSIONS & CO-POLLUTANTS IN DACS AND AVOID DISPROPORTIONATE BURDENS ON DACS

A. THE PROPOSED RNG & HYDROGEN PROJECTS WILL NOT RESULT IN MEANINGFUL EMISSIONS REDUCTIONS

Q. Will the proposed RNG interconnections or HyGrid Project reduce GHG or co-pollutant emissions in DACs?

A. No. As discussed above, replacing fossil gas with RNG and/or hydrogen will not achieve real GHG reductions, and will not reduce co-pollutants associated with end-use combustion. In fact, combustion of RNG and hydrogen could pose additional health concerns, particularly given that hydrogen combustion emits such high levels of NOx. Because DACs are already overburdened by pollution and disproportionate negative health outcomes, the proposed investments in combustion alternative fuels will only further burden DACs by prolonging the life of a system that harms their health.

Additionally, as the Climate Action Council has recognized, New York must downsize the gas system in order to comply with the CLCPA mandates.[142] As detailed above, an unmanaged transition away from the gas system will hurt low- and moderate-income ratepayers shouldering rising per-customer costs of paying for the gas distribution

[142] Scoping Plan at 350.

system. Using RNG and hydrogen to prop up the distribution system in the short term is therefore likely to disproportionately harm low- and moderate-income ratepayers and other members of DACs. The Companies can avoid these disproportionate burdens by supporting the transition to electrification and other non-pipeline alternatives, and frontloading investments in DACs.

For these reasons, the Companies' request for recovery for the proposed investments in RNG interconnections and the HyGrid project should be denied. To the extent that these investments go forward, however, they cannot contribute to satisfying the CLCPA's requirement that at least 35–40% of investments in clean energy accrue to DACs because alternative combustion fuels do not confer a benefit; they only prolong and exacerbate existing health disparities and will likely lead to higher bills for members of DACs.

Q. Will the proposed RNG investments burden DACs in any other way?

A. Yes. The four proposed RNG interconnection sites and the Newtown Creek RNG facility are sited in or very close to DACs, and therefore, any environmental and health impacts of the facilities will disproportionately burden DACs.

KEDNY seeks recovery for a proposed RNG interconnection at the Jamaica Wastewater Treatment Plant located at 154–42 134th Street in Queens, New York.[143] The facility is approximately one block from a DAC.[144]

[143] Companies' Response to DPS Interrogatory 354, Ex. 17.

[144] *See* NYSERDA, Disadvantaged Communities Map, (hereinafter "DAC Map"); *see also* Companies' Response to DPS Interrogatory 858(2), attached as Exhibit 28.

KEDNY also seeks recovery for a proposed RNG interconnection at 2671 Arthur Kill Road in Staten Island,[145] which is within a two-mile radius of a DAC.[146]

KEDLI seeks recovery for a proposed RNG interconnection at a wastewater treatment plant located at 2 Marjorie Lane in East Rockaway.[147] The facility is approximately one block from a DAC.[148]

KEDLI also seeks recovery for a proposed RNG interconnection in Calverton Executive Park,[149] which is located in a DAC.[150]

Finally, the Newtown Creek RNG facility is located in a DAC.[151]

As discussed above, RNG projects can create local environmental impacts by incentivizing methane production, thereby increasing net emissions. Additionally, anaerobic digestion has been linked to ammonia emissions, a precursor to the dangerous pollutant particulate matter. To the extent the Companies claim that the RNG production facilities will decrease emissions in nearby communities, they have not offered any support for that proposition and the RNG developers will need to undergo the Title V air permit process to determine the potential air emissions impacts.[152]

CLCPA Section 7(3) was enacted to ensure that DACs do not continue to bear the harms of our energy system, to prevent the continued buildout of energy infrastructure in DACs, and to guarantee that the transition to a clean energy economy does not

[145] Companies' Response to WE ACT Interrogatory 35, attached as Exhibit 29.
[146] Companies' Response to DPS Interrogatory 858(2), Ex. 28.
[147] Companies' Response to DPS Interrogatory 354, Ex. 17.
[148] *See* DAC Map; *see also* Companies' Response to DPS Interrogatory 858(2), Ex. 28.
[149] Companies' Response to DPS Interrogatory 354, Ex. 17.
[150] *See* DAC Map; *see also* Companies' Response to DPS Interrogatory 858(2), Ex. 28.
[151] *See* DAC Map; *see also* Companies' Response to DPS Interrogatory 858(2), Ex. 28.
[152] *See* Companies' Response to DPS Interrogatory 354, Ex. 17.

disproportionately burden DACs. The Companies have not presented a cumulative impacts report or any analysis under Section 7(3) demonstrating that the RNG proposals will comply with Section 7(3). Placing RNG production facilities in DACs is directly contrary to Section 7(3)'s intent and should not be funded by ratepayers.

B. THE COMPANIES MUST SCALE UP NPAS AND THERMAL ENERGY NETWORKS AND SIGNIFICANTLY REDUCE LEAK-PRONE PIPE PROGRAM

Q. As you testified above, the CLCPA also requires the Commission to reduce pollution in DACs. How can the Companies reduce pollution in DACs in compliance with the CLCPA?

A: A meaningful reduction of pollution in DACs must come from the simultaneous reduction of outdoor air pollutants and improvement of indoor air quality. These reductions need to come from the building sector, the electricity generation sector, and the transportation sector because all of these sectors contribute to the air pollution in DACs.

Q. How can the Companies reduce air pollution from buildings in DACs?

A. The air pollution from buildings comes from burning fossil fuels on site for heating, domestic hot water, and cooking. The appliances that burn fossil fuels for these end uses can be replaced with electric appliances that do not emit air pollution on site. There are two core strategies for reducing fossil fuel combustion in buildings: energy efficiency and electrification.

Q. How does energy efficiency reduce fossil fuel combustion in DACs?

A. "Energy efficiency" refers to several types of improvements to a building in order to retain energy and several kinds of technologies that use less energy. In general, energy efficient building improvements and technologies provide the same kind of performance as their less efficient counterparts, but they use less energy. Less energy means less fossil fuel combustion and therefore less air pollution.

Q. How does electrification reduce fossil fuel combustion in DACs?

A. Replacing gas end uses, such as heating, hot water, and cooking, with efficient electric appliances (a process called beneficial electrification) in DACs will help reduce local air pollution, especially in cases where the electricity supply for these appliances comes from a source outside a DAC or from renewable energy generation that itself has no emissions. The Companies have testified that they are pursuing electrification as a strategy for reducing greenhouse gas emissions.[153] The Companies should specifically focus their electrification efforts in DACs in order to reduce air pollution in these overburdened communities. Electrification technologies include ground-source (geothermal) heat pumps, air-source heat pumps, electric and induction cooktops, other electric appliances, and water-source heat pumps connected to a thermal energy network.

Q. How do thermal energy networks reduce fossil fuel combustion in DACs?

A. "Thermal energy networks" refers to a network of distribution mains and service lines which carry heated and cooled water or refrigerant to and from already existing heat sources and sinks. Potential heat sources include vertical or horizontal geothermal ground loops; bodies of water, water towers, or greywater; factories, data centers, sewers, subways, and other sources of excess heat; renewable co-generation plants; ice skating

[153] KEDNY/KEDLI CLCPA Panel at 11.

rinks, large-scale refrigeration, and any other areas or buildings which need to be cooled. Potential heat sinks include most buildings which need heating in the winter; water heaters and swimming pools; the ground, air, or bodies of water; power generation; and more. Water-source heat pumps within buildings exchange thermal energy with the shared ambient temperature distribution system.

Thermal energy networks would greatly benefit disadvantaged communities because the Companies would be able to spread high up-front installation costs over many ratepayers and over a long period of time, making renewable heating and cooling more affordable. Thermal energy networks can also reduce the up-front costs associated with individual electrification. They also reduce or eliminate indoor and outdoor pollution from fossil fuels. Thermal energy networks are climate resilient because they do not involve outdoor equipment which is exposed to the elements and needs to be replaced frequently. They also add cooling, whereas many people living in DACs currently do not have access to air conditioning.

Q. What are the health benefits of energy efficiency and electrification?

A. Energy efficiency prevents energy waste, which means decreased use of fossil fuels.[154] The health benefits of reduced fossil fuel combustion from the stove, boiler, and furnace result from reduced exposure to harmful air pollutants including NOx, CO, formaldehyde, and PM2.5.[155] Reduced exposure can improve rates of respiratory and cardiovascular illness, cancer, and more.[156] Additionally, energy efficient housing means

[154] ACEEE & Physicians for Soc. Resp., *Energy Efficiency and Health* (2016), https://www.aceee.org/sites/default/files/ee-health-1008.pdf.
[155] Seals & Krasner at 8, Ex. 7.
[156] *See Energy Efficiency and Health*.

increased thermal comfort. Thermal comfort can have a positive impact on arthritis and dementia, and prevent heat and cold stress that could otherwise lead to severe illness or death.[157] Furthermore, an energy efficient home has fewer instances of mold and pests, both of which can trigger asthma and allergies.[158]

The health benefits of electrification are similar to energy efficiency because combustion of fossil fuels in the home has been eliminated. Last year, WE ACT for Environmental Justice ran a study on the air quality benefits of switching gas combustion stoves to induction cooking in New York City Housing Authority ("NYCHA") households.[159] We installed induction stoves in 10 apartments, and then monitored the air quality while residents cooked and compared it to 10 control apartments (apartments using combustion gas).[160] After we completed the study, we found that induction cooking lowered daily indoor NOx concentrations by 35% compared with gas cooking.[161] It is clear that energy efficient homes which use electric appliances, such as induction stoves, are healthier for indoor environments compared to gas cooking.[162]

Q. Can you explain what Non-Pipeline Alternatives are?

A. Non-Pipeline Alternatives ("NPAs") are critical for avoiding unnecessary gas system investments. NPAs in the gas sector are the equivalent of the electric sector's "non-wires alternatives" and refer to activities or investments that delay, reduce, or avoid the need to

[157] *See* Jessel et al. at 11–12. Ex. 2.
[158] *Id.* at 8–9.
[159] Emily Pontecorvo, *A Housing Heavyweight Goes Shopping for 10,000 Induction Stoves*, Heatmap (Jul. 24, 2023), https://heatmap.news/economy/induction-stoves-new-york-city-public-housing-authority#.
[160] *Id*.
[161] *Id.*
[162] *Id*.

build or upgrade traditional gas system infrastructure such as pipelines, storage, and peaking resources.[163] NPA solutions are composed of a variety of strategies, programs, and technologies on both the demand side and supply side, including demand response, energy efficiency, electrification, thermal storage, behavioral changes, liquefied natural gas ("LNG") peaking storage, and mobile pipeline injection.[164] In practice, NPAs are typically a portfolio of solutions that offset pipeline gas demand.

Q. What are the benefits of NPAs?

A. The three main benefits of NPAs are cost reduction, risk reduction, and emissions reductions, particularly reducing or eliminating gas combustion and co-pollutants in DACs. NPAs can be less expensive than traditional utility capital investments such as pipeline replacements,[165] leading to cost savings for ratepayers and more available funds to be used for decarbonization methods. Because of these benefits, I would like to see the Companies develop and commit to a program in the Joint Proposal that strategically identifies and implements NPAs with the express purpose of leading to pollution reduction in DACs over the course of this rate term. The program should include targets and benchmarks showing that it aligns with the CLCPA mandate that 35–40% of benefits go to disadvantaged communities.

Q. What guidance has the Commission provided regarding the use of NPAs?

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[163] Advanced Energy Econ., *Non-Pipeline Alternatives (NPAs)*, (2022), https://info.aee.net/hubfs/Sarah%20S%20uploads/NPAs.pdf.

[164] *Id*.

[165] *See* Stephen Nadel, ACEEE, *Impact of Electrification and Decarbonization on Gas Distribution Costs* 27, 29, 31 (June 2023), https://www.aceee.org/sites/default/files/pdfs/U2302.pdf.

A. In March 2020, the Commission commenced a generic gas planning proceeding in Case 20-G-0131, which seeks to ensure "temporary supply, energy efficiency, electrification, and clean demand response" to "reduce or eliminate the need for gas infrastructure and investments."[166] The Gas Planning Order calls for long-term plans that consider energy efficiency and NPAs and directs utilities to include an NPA-only scenario, unless the utility can demonstrate, with sufficient evidence, that an NPA-only scenario is not feasible.[167]

Q. Have the Companies proposed a framework for consideration of NPAs?

A. The Companies, on August 10, 2022, along with other New York State gas distribution utilities, submitted a filing in the Gas Planning Proceeding (Dkt No. 20-G-0131) and proposed incentives and cost recovery mechanisms for NPA projects.[168] The filing is still under review with the Commission. On the same day, August 10, 2022, the Companies also filed an NPA screening and Suitability Criteria in the Gas Planning Proceeding. Examples of projects that may be suitable for NPA consideration are "gas distribution projects associated with load growth, and main or service replacements. Capital projects associated with immediate system needs related to safety, reliability, and service obligation, where construction is expected to commence and be completed within 24 months, would be excluded, as would any non-distribution projects where NPAs are not

[166] Case No. 20-G-0131, Order Instituting Proceeding at 7, Proceeding on Motion of the Commission in Regard to Gas Planning Procedures, N.Y. Dep't of Pub. Serv. (Mar. 19, 2020) (Dkt. No. 1).

[167] Case No. 20-G-0131, Order Adopting Gas System Planning Process at 37, Proceeding on Motion of the Commission in Regard to Gas Planning Procedures, N.Y. Dep't of Pub. Serv. (May 12, 2022) (Dkt. No. 99).

[168] Companies' Response to WE ACT Interrogatory 13, attached as Exhibit 30.

applicable."[169] Once a project is deemed eligible to proceed for NPA consideration, the Companies will evaluate the costs, size of the load relief needed, and available timeline, among other factors, to determine the feasibility of the proceeding.[170]

The Companies also contend they have limited availability to directly incentivize electrifications, which is why they work to implement NPAs.[171] As part of the Joint Proposal in the 2019 KEDNY and KEDLI Rate cases, the "Companies committed to a process by which they would refer prospective gas customers who were potential candidates for electrification to Consolidated Edison […] and PSE&G Long Island and LIPA […] as part of those companies' heat pump program."[172] The Companies are also working to implement NPAs which have the potential to enable targeted electrification as an alternative to leak-prone pipe ("LPP") replacement.[173]

Q. Do you have any concerns with the Companies' NPA Program?

A. Yes. I commend the Companies for conducting evaluations of possible NPAs before proceeding with the construction of new or replacement gas transmission and distribution infrastructure. I also commend the Companies' commitment, which was set forth in the 2019 Rate Case, to identify at least five LPP projects annually that could be replaced with NPAs, subject to customer consent.[174] However, I have four concerns with the

[169] Case No. 20-G-0131, National Grid's Proposals for Non-Pipe Alternative Screening and Suitability Criteria at 2–3, Proceeding on Motion of the Commission in Regard to Gas Planning Procedures, N.Y. Dep't of Pub. Serv. (Aug. 10, 2022) (Dkt. No. 113).
[170] *Id.* at 3.
[171] *See* Case Nos. 22-G-0226 & 22-G-0225, CLCPA Filing Package at 13, N.Y. Dep't of Pub. Serv. (Apr. 28, 2023) (Dkt. No. 2) ("KEDLI/KEDNY CLCPA Panel Testimony").
[172] *Id*. at 13.
[173] *Id*. at 14.
[174] KEDNY GIOP at 32; KEDLI GIOP at 32; Companies' Response to WE ACT Interrogatory 124, attached as Exhibit 31; Companies' Response to WE ACT Interrogatory 125, attached as Exhibit 32.

Companies' NPA criteria and program: (1) the Companies' capital expenditures do not reflect a significant use of NPAs, (2) the Companies have failed to conduct a robust comparison between NPAs and pipe replacement programs, (3) the Companies consider and refer to the HyGrid Hydrogen project as an NPA,[175] and (4) the Companies do not include DACs in the screening or suitability criteria for NPAs.[176]

Q. Can you explain, in more detail, your concerns that the Companies expenditures do not reflect a significant use of NPAs?

A. In the updated Gas Infrastructure and Operations Panel Testimonies, the Companies stated KEDNY will remove at minimum, 186 miles of LPP and KEDLI will remove at minimum, 476 miles of LPP over the four-year rate term.[177] The Companies are requesting over $2 billion in recovery for their leak-prone pipe replacement program.[178] Over the same period, the Companies have not even designated a line item of their capital spending on NPAs, since there is no commitment to the actual construction of an NPA.[179] The Companies' recent filings in the Gas Planning Proceeding, related to NPA criteria, do not suggest an intention to scale up or identify additional NPA projects.

[175] KEDLI GIOP at 74 of Exhibit __ (GIOP-5); Companies' Response to DPS Interrogatory 693, Ex. 26.

[176] Case No. 20-G-0131, National Grid's Proposals for Non-Pipe Alternative Screening and Suitability Criteria at 2–3.

[177] Case Nos. 22-G-0226 & & 22-G-0225, Corrections and Updates Testimony of Gas Infrastructure and Operations Panel at 14, 15 (June 30, 2023).

[178] In the KEDNY service territory, the Proactive Main Replacement program over the four years, will cost $931.475 million. KEDNY GIOP at 32. In KEDLI's service territory, the Proactive Main Replacement program over the four years, will cost $1,166.714 million. KEDLI GIOP at 32.

[179] Companies' Response to WE ACT Interrogatory 126, attached as Exhibit 33.

Q. You stated the Companies neglected to conduct a robust cost comparison between their NPA and leak-prone pipe replacement program. Can you please explain your concerns?

A. Yes. The Companies have a narrow view of how to achieve emissions reductions with a reasonable cost to their ratepayers. The Companies explain that the leak-prone pipe retirement and leak management programs will directly reduce emissions from the Companies' gas network.[180] Unfortunately, the Companies have not conducted a cost comparison of emissions reductions ($/MT CO2e) nor a comparison of air quality benefits between leak-prone pipe retirement and NPAs.[181] Without this necessary data to conduct a comparative analysis between LPP and proposed NPAs, the Companies are unable to conclude whether their LPP replacement program is justified as a prudent investment for the ratepayers.

Q. Do you have any other concerns?

A. Yes. I am also concerned with the Companies' failure to complete a thorough and up to date societal benefit-cost analysis ("BCA") when comparing electrification and NPA costs with LPP and alternative combustion fuels, such as the HyGrid Project located in the KEDLI service territory. The Companies have not updated their electrification cost estimates since April 2021, which were reflected in their *Pathways to Carbon-Neutral NYC Report*.[182] The Companies do not believe electrification costs are covered under the Inflation Reduction Act ("IRA") and also assess that available state rebates and

[180] KEDNY GIOP at 17.
[181] Companies' Response to NRDC Interrogatory 20, attached as Exhibit 34; Companies' Response to NRDC Interrogatory 21, attached as Exhibit 35.
[182] Companies' Response to WE ACT Interrogatory 64, attached as Exhibit 36.

incentives can change over time, making it difficult to assert their potential impact in a cost comparison.[183] Had the Companies researched rebates and incentives under the IRA for electrification, they would have found several programs, such as (1) the Home Electrification and Appliance Rebate Program,[184] (2) the Green and Resilient Retrofit Program,[185] and (3) the Residential Clean Energy Credit.[186] Even more, the Commission has approved BCAs with federal incentives, reflecting benefits used in the past such as in Case 18-E-0138, in which the Commission issued an order approving a make-ready electric vehicle program based on a New York societal cost BCA that included Federal Tax Credits as a benefit from a societal perspective. When comparing the costs of traditional infrastructure replacements and alternative combustion fuel projects with NPAs and electrification, the Companies must be conducting and analyzing a societal BCA that includes the benefits of both state and federally funded incentives.

Q. Why are you concerned that the Companies are referring to the Hydrogen Pilot Project, "HyGrid," as an NPA?

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[183] *Id*.

[184] IRA § 50122; *see also Home Electrification and Appliance Rebates*, Off. of State and Cmty. Energy Programs, https://www.energy.gov/scep/home-electrification-and-appliance-rebates (last visited Aug. 29, 2023) (The program is run by the U.S. Department of Energy and state energy offices. Approximately $4.275 billion is available through 2031 in formula grants to State Energy Offices).

[185] IRA § 30002; *see also Green and Resilient Retrofit Program (GRRP)*, U.S. Dep't of Housing & Urban Dev., https://www.hud.gov/GRRP (last visited Aug. 29, 2023) (The program allocates $837.5 million through 2028 (also $60 + $42.5 million for related programs), eligible uses for the funding includes "projects that improve energy or water efficiency, enhance indoor air quality or sustainability, implement the use of zero-emission electricity generation, low-emission building materials or processes, energy storage, or building electrification strategies; or address climate resilience").

[186] IRA § 13302 (modifying 26 U.S.C. § 25D). This program includes battery storage and residential roof top solar.

A. My understanding of an NPA is a project or program that defers, avoids, or eliminates the need for any specific, planned, or traditional fossil fuel infrastructure project on the Companies' distribution system. The Companies are requesting $13.291 million in capital costs for the HyGrid project, which will enable hydrogen blending as part of an existing green hydrogen facility in the Town of Hempstead, NY.[187] The Companies plan to inject the hydrogen produced in the HyGrid facility into the existing gas distribution system.[188] The project is planned to be operational in March 2025.[189] This Pilot Project will still require traditional fossil fuel infrastructure to transport and distribute energy to customers. The Companies should be prohibited from receiving any NPA incentives for this Pilot Project.

As I discussed in Section IV, hydrogen has a far greater risk of leakage due to its smaller molecule size compared to natural gas.[190] A recent California Public Utilities Commission study shows that the greater the hydrogen concentration in the gas network, the more significant leaks become.[191] Hydrogen leakage can reduce the alternative combustion fuel's climate benefits, as hydrogen acts as an indirect greenhouse gas in the atmosphere, increasing the amounts of other gases such as methane and reducing some of the near-term emissions reductions gained by a switch to green hydrogen.[192]

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[187] KEDLI GIOP at 80–81.
[188] *Id.*
[189] *Id.*
[190] *See* CPUC Report at 12; *see also* Christopher Findlay, *What's Your Purpose? Reusing Gas Infrastructure for Hydrogen Transportation*, Siemens Energy (Sept. 11, 2020), https://www.siemens-energy.com/global/en/news/magazine/2020/repurposing-natural-gas-infrastructure-for-hydrogen.html#:~:text=%E2%80%9CA%20changeover%20of%20the%20existing,Gascade%20CEO%20von%20dem%20Bussche.
[191] CPUC Report at 3, 11, 33.
[192] *See* Ocko & Hamburg; Sand et al.

I am also concerned with the Companies proposal to combust hydrogen in customers' homes. Hydrogen combustion's most significant public health threat is NOx, which as discussed above is a pollutant that damages heart and respiratory function, impairs lung growth in children, and leads to higher rates of emergency room visits and even premature deaths.[193] The Companies' NPA projects should be avoiding costs for traditional infrastructure and reducing emissions and pollutants that contribute to climate change and poor public health.

Q. Why should DACs be included in the criteria for identifying and selecting NPAs?

A. As explained above, the Companies assert they will identify, at minimum, ten LPP programs that will be replaced by NPAs, yet the Companies do not inquire as to whether the LPP or NPA candidates are in DACs, where members of those communities face the most pollution burdens.[194] Because DACs are not included in the Companies' LPP nor the NPAs criteria, DACs are not prioritized. In fact, for fiscal years 2023–2025, only four of the thirty proposed NPA projects identified by KEDNY and KEDLI are located in a DACs.[195] NPAs have the potential to greatly reduce pollution by reducing gas combustion at end uses, like in gas stoves and furnaces in people's homes. By electrifying end uses and switching out main and leak prone pipe replacements with NPAs, the Companies could successfully pursue a pollution reduction strategy, and by looking for project

[193] Saadat & Gersen at 18.

[194] *See* Companies' Response to DPS Interrogatory 985, attached as Exhibit 37. The Companies' system used for tracking leak-prone pipe inventories is not currently configured to identify proposed main replacement projects in Environmental Justice or Disadvantaged Communities.

[195] Companies' Response to WE ACT Interrogatory 128, attached as Exhibit 38.

candidates within DACs, the Companies could also have a positive impact on the air quality and, as a result, the health and wellbeing of the communities they serve.

In addition, the Companies assert that the gas network infrastructure investments proposed in this case, including LPP retirement, will lead to improved infrastructure in DACs, and provide multiple benefits.[196] Yet the Companies have never performed or commissioned a study to analyze the challenges of electrification in DACs, compared to other communities. Despite asserting multiple benefits, the Companies have failed to analyze the long-term effect of the maintenance of gas infrastructure on the affordability of gas for ratepayers residing in DACs, where electrification may be more slowly implemented.

Q. Do you have any recommendations for the Companies' NPA program?

A. Yes. Due to the high capital costs of the Companies' LPP program, I recommend the Companies adopt a general policy that prioritizes infrastructure repair over replacement, as well as significantly expand their NPA programs. By prioritizing pipeline patching and repairs of gas assets, the Companies would avoid installing new gas infrastructure that will inevitably become stranded. Repairs can safely extend the serviceable lives of gas assets to match the expected usage periods of those assets and reduce the rate base.[197] The Companies should (1) include DACs in the LPP algorithm and NPA criteria, (2) commission an analysis which will compare the air quality benefits of its proposed

[196] KEDLI/KEDNY CLCPA Panel Testimony at 48–49.

[197] *See* Bob Ackley & Nathan Phillips, D.C., Dep't of Energy & Env't, *Strategic Electrification in Washington, D.C.: Neighborhood Case Studies of Transition From Gas to Electric-based Building Heating* (Dec. 14, 2022), https://edocket.dcpsc.org/apis/api/Filing/download?attachId=186471&guidFileName=a9254ec8-d08f-46ed-af0e-31b28d707139.pdf.

infrastructure investments with NPAs and pipeline repair and retirement, (3) commission an analysis which will compare the cost of emissions reductions ($/MT CO2e) from the LPP retirement program with alternatives, such as targeted electrification/NPAs and pipeline retirement and repairs, and (4) commission a study which will analyze the long-term effect of the maintenance of gas infrastructure on the affordability of gas for those in DACs, where electrification may be more slowly implemented.

Q. You briefly discussed thermal energy networks above. Are you familiar with the Utility Thermal Energy Network and Jobs Act? If so, please explain.

A. Yes. In July 2022, Governor Hochul signed the Utility Thermal Energy Network and Jobs Act ("UTENJA") which seeks to remove legal barriers to utility development of thermal energy networks and enhance gas utility worker job opportunities as part of the State's decarbonization of the gas system. UTENJA directed the Commission to develop a regulatory structure for utility thermal networks that scales affordable and accessible building electrification, protects customers, and balances the role of utilities with other market and public entities.[198] In September 2022, in Case No. 22-M-0429, the Commission directed the utilities, including the Companies, to submit at least one and up to five pilot thermal energy network project proposals, with at least one in a DAC.[199]

Q. To date, how many pilot projects have the Companies proposed?

A. On January 9, 2023, the Companies submitted one thermal energy network pilot project in KEDNY's service territory and one in KEDLI's service territory.[200]

[198] KEDNY GIOP at 84.
[199] *Id*.
[200] *See* Case No. 22-M-0429, Proceeding to Implement the Requirements of the Utility Thermal Energy Network and Jobs Act, N.Y. Dep't of Pub. Serv. (Jan. 9, 2023).

Q. Are either of the pilot projects located in a disadvantaged community?

A. I commend the Companies for piloting two thermal energy network projects located in DACs. In KEDNY's service territory, National Grid has identified several NYCHA buildings on Vandalia Avenue in Brooklyn as the anchor customer for the pilot.[201] The project will serve two NYCHA 10-story apartment buildings and a low-rise community center.[202] However, the pilot program in KEDLI's service territory is located in a DAC but will not serve DAC ratepayers. Rather, KEDLI's pilot program will be on a university campus and will power a wellness center, dining hall, a high-rise dormitory, and several townhomes.[203]

Q. Are the Companies requesting cost recovery in this rate proceeding for the two thermal energy network pilot projects?

A. Not entirely. The Companies are seeking approval for cost recovery in the Thermal Energy Networks Docket, 22-M-0429. The Companies are seeking cost recovery for ten FTE positions (five for KEDLI's pilot and five for KEDNY's pilot).[204]

Q. Have the Companies committed to hiring contractors or full-time employees from DACs for the Thermal Energy Pilot Projects?

A. The Companies see apprenticeship and job training programs as key enablers for clean energy technologies, such as thermal energy networks.[205] The Companies have a "Growing NY's Green Businesses and Jobs" program called the "Grid Collective" which

[201] *Id*. Dkt. No. 22 at 6.
[202] *Id.*
[203] *Id*. Dkt. No. 21 at 8–9.
[204] KEDNY GIOP at 85; KEDLI GIOP at 85.
[205] *See* Companies' Response to WE ACT Interrogatory 133, attached as Exhibit 39; Companies' Response to WE ACT Interrogatory 134, attached as Exhibit 40.

focuses on providing clean energy training to local, low- and moderate-income communities.[206] The Grid Collective program is specifically focused on attracting historically marginalized, underserved and systemically disadvantaged (i) local businesses and (ii) local workforce (women, youth, and racial and ethnic minorities), to be identified, recruited, and trained; and then receive project/work experience; and to provide weatherization and energy efficiency training for real property types located in disadvantaged local communities.[207] The first of the Grid Collective programs launched in Jamaica, Queens in March 2023. The Companies have not extended this training for their UTENJA pilot programs. However, I am pleased to learn that the Companies are creating clean energy job trainings, apprenticeship programs, and job opportunities for low-income residents and residents of DACs.

Q. Do you have any recommendations for the Companies' Thermal Energy Network Pilots?

A. I would like to see the Companies propose additional job training and apprenticeship programs for Thermal Energy Networks. I would also recommend that the Companies track and record the trainings they offered, the location of the trainings (times and days), enrollment, graduation rates, and hiring rates. Additionally, the Commission's Order in the Thermal Energy Network Proceeding directed the utilities to develop one to five pilots. I would like to see each operating company commit to more than just the bare minimum.

[206] *Id.*
[207] *Id.*

VI. THE COMPANIES MUST ENSURE THAT AT LEAST 35% OF BENEFITS OF CLEAN ENERGY AND ENERGY EFFICIENCY SPENDING ACCRUE TO DACS

Q. What does the CLCPA require in terms of benefiting DACs with investments?

A. Per § 75-0117, the CLCPA requires:

> State agencies, authorities and entities, in consultation with the environmental justice working group and the climate action council, shall, to the extent practicable, invest or direct available and relevant programmatic resources in a manner designed to achieve a goal for disadvantaged communities to receive forty percent of overall benefits of spending on clean energy and energy efficiency programs, projects or investments in the areas of housing, workforce development, pollution reduction, low income energy assistance, energy, transportation and economic development, provided however, that disadvantaged communities shall receive no less than thirty-five percent of the overall benefits of spending on clean energy and energy efficiency programs, projects or investments and provided further that this section shall not alter funds already contracted or committed as of the effective date of this section.[208]

Q. What should be counted as a "benefit?"

A. The White House Environmental Justice Advisory Council ("WHEJAC") issued recommendations for defining "investments" and "benefits" that must be adopted. The recommendations for benefits are as follows:

> **Recommendation:** Direct Investments in Geography: Investments in defined frontline geographies (by census tract or other designations based on EJ criteria such as Minneapolis Green Zones).
>
> **Recommendation:** Direct Investments in People: Investments that benefits to Black, Indigenous, Latinx, Asian, Pacific Islander, GLBTQ [sic], People of Color, and Immigrants to improve health and economic opportunities.
>
> **Recommendation:** Indirect and direct Investment in Community: Investment which support local communities, community-based organizations, community ownership, cooperatives, small-business, community job training and local ownership tracks, etc.

[208] ECL § 75-0117.

> **Recommendation:** External Direct Investment benefits: Investments that are outside EJ communities but provide essential services to EJ such as water, energy, and sanitation.
>
> **Recommendation:** All Investments: Must do no harm to EJ Communities.[209]

The benefits of the transition to a renewable energy-based economy are numerous. They include, but are not limited to, health improvements from cleaner air, job creation, investment and wealth-building opportunities, business development opportunities, energy resilience, and utility bill reductions.

Q. What are the health benefits and how should they be quantified?

A. As described above, the burning of fossil fuel for electricity, direct use in buildings, and transportation has a detrimental impact on the health of the Companies' customers and even more so in DACs, which by definition bear the cumulative impacts of multiple pollution sources. Reducing the local air pollution from fossil fuel combustion benefits people's health, and by extension, reduced healthcare costs, which have a financial benefit. Direct health benefits include reduction of air pollution-related illness and increased thermal comfort. Health benefits can be quantified by the reduction in hospitalizations, emergency room visits for air pollution-related illness such as asthma, heat or cold stress,[210] the number of prevented potential years of life lost, and increased life expectancy.[211] They can also be calculated through a reduction in the health impacts

[209] White House Environmental Justice Advisory Council ("WHEJAC"), *Final Recommendations: Justice40 Climate and Economic Justice Screening Tool & Executive Order 12898 Revisions* 57 (2021), https://www.epa.gov/sites/default/files/2021-05/documents/whiteh2.pdf.

[210] *See* Sarunya Sujaritpong et al., *Quantifying the Health Impacts of Air Pollution Under a Changing Climate—a Review of Approaches and Methodology*, 58 Int'l J. Biometeorology 149 (2013), attached as Exhibit 41.

[211] *See id*. at 151.

from thermal discomfort and air pollution, such as lower rates of preterm birth and low birthweight. Furthermore, increased days at work or school due to improved health and decreased restricted activity days can be modeled using existing cohort and time-scale studies.[212] The DAC criteria has a list of health-related indicators that can be tracked for improvement.

Q. What are the job creation benefits and how should they be quantified?

A. According to the Scoping Plan, the transition to a renewables-based economy will result in the creation of 269,000 jobs between now and 2050.[213] To the extent that the Companies are hiring workers or contracting with other companies that hire workers, the Companies should ensure that workers from DACs are given appropriate opportunities. Employing members of DACs will benefit those individuals as well as their communities as a whole.

The Companies need to look at their own hiring practices first and foremost. On June 16, 2022, the Commission launched a Utility Employee Diversity, Equity and Inclusion ("DEI") Initiative.[214] The Commission directed the utilities, including the Companies, to develop and file DEI plans. On June 1, 2023, National Grid filed its first DEI Strategic Plan.[215] Of the new positions related to its clean energy programs, the Companies need to make a concerted effort to hire 40% of new workers from DACs and should track and report this effort and the results to the Commission.

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[212] *See* EPA, *Quantifying the Emissions and Health Benefits of Energy Efficiency and Renewable Energy* at 4-27 (2018), https://www.epa.gov/sites/default/files/2018-07/documents/mbg_2-4_emissionshealthbenefits.pdf.

[213] Scoping Plan at 90.

[214] *See* Case No. 22-M-0314, Proceeding to Review Utilities' Diversity, Equity, and Inclusion Practices, N.Y. Dep't of Pub. Serv. (June 16, 2022).

[215] *Id*. at Dkt. No. 11.

The Companies should also consider their contractors and whether these contractors hire from DACs. The Companies currently track vendors and contractors by (1) minority-owned, (2) women-owned, and (3) veteran-owned business but do not track this information by individual operating company.[216] The Companies should continue hiring contractors and vendors from minority-, women-, and veteran-owned businesses as well as from DACs when choosing contractors and vendors and make this the desired criteria. The Companies should also track this information by individual operating company. Similarly, the Companies should track and report the hiring practices and workforce demographics of their contractors with the goal of 40% of the workforce of their contractors on new investments and programs being from DACs.

The Companies should also ensure that the people hired from DACs are considered for and hired into upper-level positions, paid fair wages, and are offered training opportunities and full-time work. For these jobs to be counted as a benefit for the purposes of CLCPA compliance, they must lead to long-term financial security.

Q. What are the investment, wealth-building, and economic development opportunities and how should they be quantified?

A. The development of a renewable energy-based economy will provide opportunities for new businesses, new investments, and new economic development. To date, people of color and low-income individuals are underrepresented in the clean energy industry, both in the workforce and in financial benefits. The 2019 Solar Energy Industries Association

[216] Companies' Response to WE ACT Interrogatory 66, attached as Exhibit 42.

reported that Black/African American and Latino/a workers make up 7.6% and 16.9% of the overall industry respectively,[217] while 88% of executives identify as white.[218]

According to a study by the American Association of Blacks in Energy, "while Black households spent $41 billion on energy in 2009, they held only 1.1% of energy jobs and gained only 0.01% of the revenue from energy sector profits."[219] These stark disparities point to the need for communities of color to build ownership within the clean energy industry. Investments must lead to wealth-building and financial security, and to achieve this long-term goal, it is necessary to invest in ownership opportunities and ensure members of DACs are in upper-level positions in businesses.

For example, WE ACT for Environmental Justice trained over 100 people in solar installation. Our evaluation found that even years after the training, these trainees, all who are from areas now characterized as DACs, were not employed in clean energy jobs. To achieve job security and community ownership of renewable energy, they created a solar worker cooperative named Solar Uptown Now Services. Since its launch in 2019, the cooperative has completed several solar projects, including a 4 MW installation.[220] The Companies can invest in a workforce that is from DACs, who are able to build ownership of renewable energy projects in their neighborhoods. Methods to quantify the opportunities offered include wage levels, level of position in clean energy companies,

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[217] Solar Energy Indus. Ass'n & the Solar Found., *U.S. Solar Industry Diversity Study 2019* at 7 (2019), https://www.seia.org/sites/default/files/2019-05/Solar-Industry-Diversity-Study-2019.pdf.

[218] *Id*. at 28.

[219] Ariel Drehobl et al., Am. Council for an Energy-Efficient Econ., *How High Are Household Energy Burdens?* 3 (2020), attached as Exhibit 43.

[220] *Solar Uptown Now Services Solar Workers Cooperative*, WE ACT for Env't Just., https://www.weact.org/campaigns/solar-uptown-now-services-solar-workers-cooperative/ (last visited Aug. 30, 2023).

job training statistics by geography and race, ownership or investment in clean energy companies and technologies, existence of microgrids and community solar, and increasing clean energy businesses rooted in DACs.

Q. How are the Companies doing when it comes to achieving the CLCPA requirement that at least 35% of benefits accrue to DACs?

A. The Companies do not know and do not have the information relating to the total program expenditures or program expenditures for low-income and non-low-income customers located in DACs for 2023–2027, for programs related to: reducing leaks in pipes, reducing local GHG emissions, improving air quality, and enhancing safety and reliability of service to customers.[221] The Companies simply state a "large portion" of capital investments will "upgrade facilities" in DACs.[222] The Companies should be tracking and quantifying the capital expenditures and programs that will be benefited by DACs.

To highlight my concern, the Companies are planning to upgrade their Advanced Metering Infrastructure including the (1) Automatic Meter Reading Replacement Program, (2) Advanced Communication Infrastructure and Residential Methane Detector Programs, and (3) the AMI Network Program[223] but will not include DACs in the algorithms they are currently developing.[224] In not including DACs in the algorithm, the

[221] Companies' Response to DPS Interrogatory 759, attached as Exhibit 44; Companies' Response to DPS Interrogatory 760, attached as Exhibit 45.
[222] *Id.*
[223] *See* KEDNY GIOP at 4–18 of Exhibit __ (GIOP-5).
[224] Companies' Response to WE ACT Interrogatory 37, attached as Exhibit 46; Companies' Response to WE ACT Interrogatory 39, attached as Exhibit 47; Companies' Response to WE ACT Interrogatory 40, attached as Exhibit 48.

Companies cannot ensure customers residing in DACs will receive 35–40% of the benefits of reducing emissions and of safer and more reliable infrastructure.

Q. Even though the Companies are gas only utilities, are they engaged in any electrification programs?

A. As part of the Joint Proposal in the 2019 KEDNY and KEDLI Rate Cases, the Companies committed to a process by which they would refer prospective gas customers who were potential candidates for heat pump electrification programs to duel fuel utilities such as Consolidated Edison and PSE&G Long Island and LIPA.[225] The Companies explained they referred nearly 800 potential customers to their peer utilities in 2022.[226] Unfortunately, the Companies cannot confirm how many of those customers took advantage of heat pump programs but can confirm that 281 customers have either moved forward with the installation of new or upgraded natural gas service, or are in the process of having the gas service installed.[227]

Q. How many of the 800 potential customers referred to electric heat pump programs were in DACs?

A. The Companies do not track whether callers or potential customers are located in DACs.[228]

Q. Are you familiar with the Companies' Weatherization, Health and Safety Pilot Program?

A. Yes

[225] KEDLI/KEDNY CLCPA Panel Testimony at 13.
[226] *Id*.
[227] Companies' Response to WE ACT Interrogatory 68, attached as Exhibit 49.
[228] *Id*.

Q. Can you please explain the Pilot Program?

A. The Companies have developed a residential weatherization program where customers work with an approved program partner, a weatherization contractor who participates in the program, to identify and install weatherization home upgrades.[229] The Companies have also developed and launched weatherization programs to address the needs of commercial and industrial and multifamily customers.[230] All of the Companies' weatherization programs offer incentives on weatherization upgrades including wall, attic, and floor insulation, air sealing, window upgrades, and pipe insulation.[231]

Q. Do you know the annual expenditures for the Pilot Program for each year, including outreach and education, marketing, incentives/projects, third-party contractors, or any other relevant costs?

A. Yes. In 2022 and up until March 2023, KEDLI spent $1,477,958 and KEDNY spent $267,262.[232]

Q. Do you know how much of the Companies' annual expenditures for the Pilot Program were spent on DACs?

A. In KEDLI's service territory, the Company spent 95% in 2022, and 100% in 2023, on DACs.[233] In KEDNY's service territory, the Company spent 100% of total expenditures on DACs and in 2023, the Company has not expended any funds.[234]

[229] Companies' Response to WE ACT Interrogatory 70, attached as Exhibit 50.
[230] *Id.*
[231] *Id.*
[232] Companies' Response to DPS Interrogatory 491, attached as Exhibit 51.
[233] Companies' Response to WE ACT Interrogatory 129, attached as Exhibit 52.
[234] Companies' Response to WE ACT Interrogatory 130, attached as Exhibit 53.

Q. How is the Company ensuring that this Pilot Program is reaching members of disadvantaged communities?

A. The Companies are spending the majority of the allotted funds on members of disadvantaged communities and for that, I am pleased. However, I am concerned that the Companies or its contractors are wrongly deeming potential candidates as ineligible.[235] According to the Companies, "health and safety issues on the premises"[236] will disqualify a candidate's eligibility, such as:

- Moisture on the premises;
- Carbon monoxide hazards (venting issues or stove/oven in need of replacement);
- Gas leaks and pipe repair;
- Inoperable, inefficient bathroom fans;
- Ventilation systems in need of replacement;
- Needed roof repair or replacement;
- Septic and plumbing issues;
- Suspected environmental hazards (asbestos, vermiculate, mold and mildew); and
- Suspected electrical issues.[237]

The Commission should require the Companies to track and report the dwellings that were deemed ineligible.[238]

Q. Why is the Weatherization, Health and Safety Pilot Program so important for customers residing in DACs?

[235] *See* Companies' Response to DPS Interrogatory 491, Ex. 51.
[236] *Id*.
[237] Companies' Response to WE ACT Interrogatory 96, attached as Exhibit 54.
[238] The Companies should also follow standards and quality assurance guidelines set forth by the U.S. Department of Energy. *See Weatherization Program Notice 22-7: Weatherization Health and Safety*, U.S. Dep't of Energy, https://www.energy.gov/scep/wap/articles/weatherization-program-notice-22-7-weatherization-health-and-safety (last visited Sept. 1, 2023).

A. The Companies state the primary goal of the program is to remove the health and safety barriers present at low-income customers' premises and those living in DACs.[239] This Program, the Companies continue, will help increase engagement in and awareness of energy efficiency programs and will enable coordination with NYSERDA's Clean Energy Hubs.[240]

This Pilot Program will increase the energy efficiency of residential homes owned or occupied by low-income ratepayers, will reduce their total energy burdens, and improve their health and safety, especially for those that are particularly vulnerable, such as the elderly, the disabled, and children. This Pilot Program, as well as other weatherization programs also produce new, competitive jobs and install emerging technologies, all while helping the most vulnerable families in parts of Brooklyn and Long Island.

Q. Can you explain what the Residential Weatherization Program is?

A. The Companies have developed a residential weatherization program where customers work with a Company-approved program partner, a weatherization contractor who participates in the program that identifies potential weatherization home upgrades.[241] To educate customers, the Companies will distribute flyers and work with contractors to network and education other contractors about the programs.[242]

Q. Are the Companies ensuring that the Residential Weatherization Program is reaching members of disadvantaged communities?

[239] Companies' Response to DPS Interrogatory 715, attached as Exhibit 55.
[240] *Id*.
[241] Companies' Response to WE ACT Interrogatory 70, Ex. 50.
[242] *Id*.

A. The Companies are not ensuring that the Residential Weatherization Program is reaching DACs. In KEDLI's service territory, the Company focused on serving low- to moderate-income customers that were identified as in need of remediation of health and safety issues through the Companies' HEAT or EmPower Program.[243] In KEDNY's service territory, the Company only shared one flyer about the program at some social gathering facilities and events.[244] The Pilot Program launched in 2021 and four residential projects were completed by the end of the year and the following year in 2022, the program completed 1,936 residential projects.[245] In the first quarter of 2023 (January through March), the Companies completed 939 residential projects.[246] Since the inception of the project, the Companies have completed 2,879 residential weatherization projects, but only 370 residential customers have resided in DACs.[247]

Q. Do the Companies have a plan to address this lack of enrollment in and outreach to DACs?

A. The Companies explain they are anticipating additional residential programs in 2023 for DACs.[248] The Companies intend to provide information about the Pilot Program to all Energy Affordability Program participants and plan to interact with customers and community organizations located in DACs to raise awareness.[249]

Q. What are your recommendations for how the Commission should remedy this in this rate case?

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[243] Companies' Response to DPS Interrogatory 491, Ex. 51.
[244] *Id*.
[245] Companies' Response to WE ACT Interrogatory 70, Ex. 50.
[246] *Id*.
[247] *Id*.
[248] *Id*.
[249] Companies' Response to DPS Interrogatory 491, Ex. 51.

A. (1) The Commission should require the Companies to develop a methodology to track, quantify, and report the health improvements from cleaner air, job creation, investment and wealth-building opportunities, business development opportunities, energy resilience, and utility bill reductions that result from the Companies' activities in DACs.

(2) The Commission should compare how the Companies' reported benefits in DACs measure compared to the 35–40% goal set by the CLCPA, and if the Companies are not meeting that goal, the Commission should require the Companies to develop a plan for reaching compliance with the CLCPA.

(3) The Companies should continue educating contractors on emerging technology and available state and federal rebates and incentives. The Companies should also expand their contractor partnerships to ensure they are working with a diverse workforce which speaks multiple languages.

VII. PAYMENTS THAT FUND ADVOCACY, POLITICAL ACTIVITY, OR MARKETING CONTRARY TO RATEPAYER INTERESTS SHOULD NOT BE RECOVERABLE

Q. Do you have any other concerns about the Companies' rate requests? If so, please explain your concerns.

A. Yes. The Companies' Historic Test Year includes expenses for advocacy and political activities and marketing to promote National Grid's interests. These expenses should not be funded by ratepayers.

Q. Do any laws place restrictions on the recoverability of political activity? If so, please explain.

A. Yes. The New York Public Service Law prohibits the Commission from approving a rate request that includes the cost of legislative lobbying or dues to any entity that participates in legislative lobbying.[250] Additionally, New York has adopted the Federal Energy Regulatory Commission's Uniform System of Accounts.[251] Under the Uniform System of Accounts, political expenditures must be recorded "below the line" as outlined in Account 426.4:

> This account shall include expenditures for the purpose of influencing public opinion with respect to the election or appointment of public officials, referenda, legislation, or ordinances (either with respect to the possible adoption of new referenda, legislation or ordinances or repeal or modification of existing referenda, legislation or ordinances) or approval, modification, or revocation of franchises; or for the purpose of influencing the decisions of public officials.[252]

Q. Has the Commission issued any guidelines concerning these types of expenses?

A. Yes. In 1977 the Commission issued a policy statement setting forth the types of advertising costs that utilities may not recover from ratepayers.[253] The Commission explained that "[i]t is generally accepted that advertising which states a utility's position on a matter of public controversy necessarily reflects the political views or self-interest of managements or shareholders and should not be borne by ratepayers" and that ratepayers should not fund any "advertising which seeks to sway opinion — legislative, environmental, governmental, consumer or any other kind — to the industry's position on

[250] *See* PSL § 114-a ("In determining rates to be charged customers, the commission shall not include the cost of legislative lobbying on behalf of any public utility as part of any such utility's operational costs and the commission shall not include the cost of membership dues for any organization, association, institution, corporation or any other entity that engages in legislative lobbying as part of any such utility's operational costs.").
[251] *See* 16 NYCRR Pt. 312.1.
[252] 18 C.F.R. Pt. 201.
[253] *See In re Advertising and Promotional Practices*, No. 27052, 1977 WL 201861 (N.Y.P.S.C. Feb. 25, 1977).

public policy disputes."[254] The 1977 policy statement also provided that only a very small percentage of a utility's operating revenues could go towards "institutional advertising," meaning advertising that is not clearly controversial or political, "for example, in which a company defends, justifies, or even merely explains and describes its activities."[255] The policy statement explained that the Commission would determine the appropriate percentage for each utility, which would likely "range between 1/10 and 1/25 of one percent of operating revenues."[256] The policy statement further stated:

> This plan will obviously not solve all the problems we have described: we will still have to determine which advertisements are clearly political and/or self-serving, and specifically disallowed; and parties in rate cases would still be free to question the propriety of the lump sum allowance for the informational and other institutional advertising categories.[257]

Q. How do the Companies track their expenses?

A. According to KEDNY and KEDLI, they track expenses at the vendor level and vendor charges are booked to internal orders based on the operating companies that benefit from the service.[258] Internal orders settle to a Federal Energy Regulatory Commission account that determines the order's classification as above or below the line for purposes of ratemaking.[259] The Companies use the SAP time entry system to track employee time.[260]

Q. Have the Companies used any ratepayer funds for political activities?

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[254] *Id*. at *4.
[255] *Id*. at *5.
[256] *Id*.
[257] *Id*.
[258] Companies' Response to WE ACT Interrogatory 87(c), attached as Exhibit 56.
[259] *Id*.
[260] Companies' Response to WE ACT Interrogatory 82(d), attached as Exhibit 57.

A. Yes. The Companies have engaged in political activities such as legislative lobbying and regulatory advocacy without recording the corresponding time or expenses below the line.

For example, in November 2021, National Grid Vice President of Corporate Affairs Bryan Grimaldi delivered testimony before the New York City Council Committee on Environmental Protection on Intro 2317, a bill to phase out fossil fuel combustion in new buildings.[261] Grimaldi's salary is included in the Company's revenue requirement, except for the portion of his salary associated with below-the-line work activity.[262] However, the time spent preparing and delivering testimony on Intro 2317 was not recorded below the line.[263] Because this testimony was for "the purpose of influencing the decisions of public officials,"[264] the associated time and expense should have been recorded below the line and not ratepayer-funded.

Similarly, in June 2022, National Grid submitted comments to the Climate Action Council on the draft Scoping Plan.[265] The time spent preparing those comments was not tracked or recorded below the line.[266] National Grid employees also testified at hearings before the Climate Action Council on the draft Scoping Plan.[267] That time was not

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[261] *National Grid Submits Testimony for the NYC Council Committee on Environmental Protection*, Nat'l Grid, (Nov. 17, 2021), https://www.nationalgridus.com/News/2021/11/National-Grid-Submits-Testimony-for-the-NYC-Council-Committee-on-Environmental-Protection-Committee-Oversight-Hearing-on-Building-Electrification/; *see also* Companies' Response to DPS Interrogatory 5 Attachment 1 at 816, attached as Exhibit 58; Companies' Response to WE ACT Interrogatory 82, Ex. 57.

[262] Companies' Response to WE ACT Interrogatory 91, attached as Exhibit 59.

[263] Companies' Response to WE ACT Interrogatory 82, Ex. 57.

[264] 18 C.F.R. Pt. 201.

[265] *See* Companies' Response to WE ACT Interrogatory 83, attached as Exhibit 60.

[266] *Id*.

[267] This testimony is available at https://climate.ny.gov/Resources/Public-Comment-Process.

recorded below the line.[268] These activities, which are for "the purpose of influencing the decisions of public officials,"[269] should be recorded below the line and not financed by ratepayers. Additionally, in June 2023, National Grid Vice President for U.S. Policy and Regulation Sandy Grace submitted a comment letter to DEC and NYSERDA about New York's proposed cap and invest program, which is being developed to implement the CLCPA.[270] Costs related to the preparation of those comments were not recorded below the line.[271]

Because the Companies did not track the time spent on these activities or record that time below the line, I am concerned that they are requesting rate recovery for activities that should be paid for by shareholders and not by ratepayers.[272]

Q. Have the Companies used any ratepayer funds for other political activities?

A. Yes. The Companies have used ratepayer funds to develop advertising and other materials promoting the Companies' views on public policy disputes, including the role of alternative combustion fuels such as RNG and hydrogen, which the Companies have then used in political advocacy.

In April 2022, National Grid released a report called "Our clean energy vision," which makes policy recommendations in support of using hydrogen and RNG to

[268] Companies' Response to WE ACT Interrogatory 135, attached as Exhibit 61.
[269] 18 C.F.R. Pt. 201.
[270] This comment letter is available at: https://www.commentmanagement.com/comment/view?key=rvk_oeyTVlYPT93izM5yuUjVycbWtysWz6SIxq1Bgv8&l=244468.
[271] Companies' Response to WE ACT Interrogatory 136, attached as Exhibit 62.
[272] While the Companies state that work on the cap and invest comments took place after the Historic Test Year and therefore associated costs are not reflected in the revenue requirements or the proposed rates at issue here, the Companies' decision to include those costs above the line is indicative of a larger trend of incorporating advocacy costs into rates.

decarbonize New York's gas supply and discourages policymakers from supporting full electrification of the building sector.[273] National Grid hosts the report on a webpage titled "Our vision for a fossil-free future."[274] The website also hosts a video describing and promoting the report, a podcast called "Clean Energy Revolution" about National Grid's vision for a "net zero world," and quotes from National Grid supporters including state and local elected officials.[275]

The Clean Energy Vision report, website, video, and podcast were developed by the Companies' internal teams whose labor costs are reflected in the Historic Test Year and revenue requirement.[276] The Companies did not track the costs associated with those specific work activities.[277] In other words, ratepayers paid for the report and the materials to promote it.

Q. How does the Clean Energy Vision report promote the Companies' position on public policy disputes?

A. The report posits that a hybrid gas-electric approach is "[a] better way to net zero" than converting entirely to zero-emission technologies such as electric heat pumps and

[273] *See generally* National Grid, *Our Clean Energy Vision* (2022) ("Clean Energy Vision Report"), https://www.nationalgrid.com/document/146251/download.
[274] *Our Vision for a Fossil-free Future*, Nat'l Grid, https://www.nationalgrid.com/us/fossilfree (last visited Aug. 30, 2023).
[275] *See id.*
[276] Companies' Response to WE ACT Interrogatory 46, attached as Exhibit 63; Companies' Response to WE ACT Interrogatory 47, attached as Exhibit 64; Companies' Response to WE ACT Interrogatory 48, attached as Exhibit 65; Companies' Response to WE ACT Interrogatory 50, attached as Exhibit 66.
[277] *Id.*

geothermal heating.[278] The report promotes RNG and hydrogen as clean, reliable, and safe fuels while painting electrification as costly and restrictive.[279]

The report presents National Grid's opinions on questions that are at the center of New York climate policy debates, including within the Climate Action Council and the Legislature. The Climate Action Council spent years debating the role of alternative combustion fuels such as hydrogen and RNG and analyzing the costs and benefits of different approaches.[280] In fact, a utility executive on the Climate Action Council voted against the final Scoping Plan precisely because the Plan prioritizes electrification and does not envision a significant role for these alternative fuels in the building sector.[281] Similarly, debate over the recently enacted Intro 2317 and All-Electric Building Act—which will require new buildings to be all-electric in New York City and State respectively—is inextricably linked with the future of the gas system.

Q. How have the Companies used the Clean Energy Vision report?

[278] Clean Energy Vision Report at 7.
[279] *Id*. at 7–10, 16.
[280] *See, e.g.*, Scoping Plan at 130 (discussing integration analysis and findings that the Accelerated Transition Away from Combustion pathway produces $130 billion in net benefits, $15 billion more than the Strategic use of Low Carbon Fuels pathway); Minutes of the Climate Action Council Meeting at 16 (Dec. 19, 2022), https://climate.ny.gov/-/media/Project/Climate/Files/CAC5December2022MeetingMinutesFinal19December2022.pdf (discussing revisions to the Gas System Transition chapter of Scoping Plan regarding RNG and hydrogen); Presentation of Climate Action Council Meeting, (Apr. 18, 2022), https://climate.ny.gov/-/media/Project/Climate/Files/2022-04-18-CAC-meeting-presentation.pdf (detailing the work of the Gas System Transition and Alternative Fuels subgroups throughout 2022).
[281] *See* Marie French, *New York Passes Sweeping Plan to Reduce Emissions and 'Lead the Way on Solving Climate Change'* Politico (Dec. 19, 2022), https://www.politico.com/news/2022/12/19/new-york-emissions-climate-change-00074600.

A. The Companies include information about and links to the report in communications with customers such as notices and emails.[282] The costs associated with these communications are reflected in the Historic Test Year.[283] In this way the Companies are promoting their positions on RNG, hydrogen, and the viability of electrification to their customers using ratepayer money.

National Grid also promotes the report on social media.[284] National Grid's social media accounts are managed by internal employees whose labor costs are reflected in the Historic Test Year and revenue requirement.[285] To assist with graphic design, reporting, video, and moderation, the Companies sometimes use the services of vendors whose costs are reflected in the Historic Test Year.[286]

Not only does the Clean Energy Vision report promote National Grid's position on live public policy questions that affect its business—the report is also an advocacy document that National Grid has used in regulatory lobbying.

For example, National Grid submitted comments to the Climate Action Council on the draft Scoping Plan that presented the utility's energy vision as a policy pathway.[287]

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[282] *See, e.g.*, Companies' Response to WE ACT Interrogatory 45 Attachment 1 at 117, attached as Exhibit 67; Companies' Response to WE ACT Interrogatory 45 Attachment 5 at 9, 11, 13, 22, 24, 26, attached as Exhibit 68; Companies' Response to WE ACT Interrogatory 45 Attachment 6 at 64, 66, 68, attached as Exhibit 69; Companies' Response to WE ACT Interrogatory 49, attached as Exhibit 70.

[283] Companies' Response to WE ACT Interrogatory 45, attached as Exhibit 71; Companies' Response to WE ACT Interrogatory 49, Ex. 70.

[284] *See, e.g.*, National Grid US, Facebook (Apr. 19, 2022), https://m.facebook.com/story.php?story_fbid=10158558621569249&substory_index=0&id=255170664248&m_entstream_source=timeline.

[285] Companies' Response to WE ACT Interrogatory 59, attached as Exhibit 72.

[286] *Id*.

[287] *See* Press Release, National Grid, *National Grid CAC Draft Scoping Plan Comments Encourage Fair and Equitable Clean Energy Goals for All* (July 8, 2022),

The comments relied heavily on the Clean Energy Vision report to put forward National Grid's preferred policy pathway.[288] In addition to paying for the time spent preparing the comments themselves, ratepayers funded the research and writing that formed the basis for the comments.[289]

In earnings reports, National Grid has been explicit that the Clean Energy Vision report is a political advocacy tool. In a November 2022 half-year results call, Chief Operating Officer John Pettigrew explained:

> [A]cross the US northeast, we're advocating working with all stakeholders in progressing our Clean Energy Vision. In December, the New York State Climate Action Council is expecting to make its policy recommendations on the role of gas networks in meeting the State's net zero targets. These recommendations will be followed by consultations, regulatory processes and legislation, and we're closely engaged in the debate.[290]

In a May 2023 full-year results call, Pettigrew stated that National Grid has "[p]rogressed our Clean Energy Vision for our US networks through continued engagement with our regulators and elected officials on our legislative and policy agenda."[291]

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https://www.nationalgridus.com/News/2022/07/National-Grid-CAC-Draft-Scoping-Plan-Comments-Encourage-Fair-and-Equitable-Clean-Energy-Goals-for-All-/.

[288] *See* National Grid, *Comments on New York State Climate Action Council's December 31, 2021, Draft Scoping Plan* at 15, 17, 19, 25 (July 1, 2022), https://acrobat.adobe.com/link/track?uri=urn%3Aaaid%3Ascds%3AUS%3A781792cc-bbff-4a60-ba09-f5497c43f340.

[289] Companies' Response to WE ACT Interrogatory 83, Ex. 60.

[290] National Grid, *Half Year Results 2022/23* at 15 (Nov. 10, 2022), https://www.nationalgrid.com/document/148631/download.

[291] National Grid, *National Grid PLC Full-Year Results 2022/23*, Fin. Times (May 18, 2023), https://markets.ft.com/data/announce/detail?dockey=1323-15961604-3FNQF8SQQ8JUU5EU925QQAUHIC.

Moreover, National Grid has engaged in lobbying on bills pending before the Legislature and New York City Council regarding energy and environmental issues.[292] National Grid also gives millions of dollars to organizations like New Yorkers for Affordable Energy that have engaged in lobbying on energy and environmental issues.[293] If National Grid or any trade association used the Clean Energy Vision report in its lobbying—as indicated by Pettigrew's statement that National Grid has "[p]rogressed our Clean Energy Vision" with elected officials to push National Grid's legislative agenda—then ratepayers indirectly paid for that political activity.

When asked why National Grid published and promoted the Clean Energy Vision report before finalizing a study on the CLCPA that the PSC had mandated, the Company responded:

> National Grid published the Clean Energy Vision report to provide a high-level overview of its vision to achieve net zero GHG emissions by 2050 and fully eliminate the use of fossil fuels for the provision of heat to customers in New York and Massachusetts. Given that policy discussions in both states on decarbonization pathways to achieve net zero was well under way,

[292] *See, e.g.*, 2021–2022 Lobbyist Registration of Dickinson & Avella, PLLC, attached as Exhibit 73, (showing lobbying on behalf of the State Energy Coalition, of which National Grid is a member, on a state bill regarding energy issues); 2021–2022 Lobbyist Registration of Yoswein New York, Inc., attached as Exhibit 74, (showing lobbying on behalf of National Grid on a bill before the New York City Council regarding environmental legislation).

[293] *See, e.g.*, Case No. 22-G-0225, Revenue Requirement Panel at Exhibit __ (RRP-2), Schedule 1 at 2, Schedule 2 at 44, 52, Brooklyn Union Gas Co., N.Y. Dep't of Pub. Serv. (Apr. 28, 2023) (Dkt. No. 2); Case No. 22-G-0226, Revenue Requirement Panel at Exhibit __ (RRP-2), Schedule 1 at 2, Schedule 2 at 44, 52, KeySpan Gas East Corp., N.Y. Dep't of Pub. Serv. (Apr. 28, 2023) (Dkt. No. 2); Companies' Response to WE ACT Interrogatory 11, attached as Exhibit 75; 2021–2022 Lobbyist Registration of New Yorkers for Affordable Energy, Inc., attached as Exhibit 76, (showing lobbying on a state bill related to energy use); 2021–2022 Lobbyist Registration of SKDKnickerbocker, LLC, attached as Exhibit 77, (showing lobbying on behalf of New Yorkers for Affordable Energy on a state bill related to energy issues).

> National Grid, like other stakeholders, developed and contributed information to inform that process.[294]

In other words, National Grid developed the Clean Energy Report to support its policy advocacy as New York implements the CLCPA. Ratepayers should not fund this type of work.

Q. Has National Grid engaged in other ratepayer-funded advocacy activities?

A. Yes. Among the ratepayer-funded outreach and education materials are communications promoting National Grid's views on climate policy issues. For example, among the materials is a notice to customers stating:

> You've made an important decision by choosing natural gas. It's the fuel of choice for home heating. The best chefs insist on it for cooking. And more than 95% of the natural gas used in the United States is produced right here in North America. That makes it a reliable energy choice, but natural gas is also the cleanest and most efficient fossil fuel. That's good for your home or business, and our planet. Using natural gas instead of other fuels reduces the emissions that contribute to smog, acid rain and global warming.[295]

Similarly, an issue of a pipeline safety newsletter for customers included information about National Grid's promotion of its "net zero" vision at the 2021 United Nations Climate Change negotiations and linked to the Clean Energy Revolution podcast discussed above.[296] In the words of the Commission's 1977 policy statement, such a notice by National Grid "seeks to sway opinion . . . to the industry's position on public policy disputes" by promoting natural gas as a clean fuel and the best choice for cooking and heating.[297] This type of communication should be financed, if at all, by shareholders, not ratepayers.

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[294] Companies' Response to NRDC Interrogatory 37(b), attached as Exhibit 78.
[295] Companies' Response to WE ACT Interrogatory 45, Attachment 5 at 41, Ex. 68.
[296] Companies' Response to WE ACT Interrogatory 45, Attachment 6 at 43, Ex. 69.
[297] *See In re Advertising and Promotional Practices*, 1977 WL 201861 at *4.

Q. Are the Companies seeking recovery for any other costs that should be unrecoverable as political?

A. Yes. Both KEDNY and KEDLI seek recovery of dues paid to trade associations.[298] In particular, KEDNY seeks recovery of $2,419.02 and KEDLI seeks recovery of $2,974 in dues to the National Insurance Crime Bureau, which has registered lobbyists at the U.S. Senate and House of Representatives.[299] The Public Service Law expressly prohibits recovery of member dues for any "entity that engages in legislative lobbying," including lobbying before Congress.[300] Accordingly, the request to recover dues to the National Insurance Crime Bureau should be denied.

VIII. RECOMMENDATIONS

A. RATEPAYERS SHOULD NOT PAY FOR RNG OR HYDROGEN PROJECTS

Q. Do you recommend approving cost recovery related to the four proposed RNG interconnection facilities?

A. No. As discussed above, to the extent that RNG is produced, it should be reserved for truly hard-to-electrify uses in certain industries. Using RNG as a replacement for fossil natural gas in the gas distribution system will only maintain or exacerbate current health harms and disparities, prolong a system that will become a stranded asset, and result in higher utility bills for low-income ratepayers who can least afford to shoulder that

[298] Companies' Response to DPS Interrogatory 48 Attachment 1, attached as Exhibit 79; Companies' Response to DPS Interrogatory 48 Attachment 2, attached as Exhibit 80.

[299] Companies' Response to DPS Interrogatory 48 Attachment 1, Ex. 79; Companies' Response to DPS Interrogatory 48 Attachment 2, Ex. 80. *See also* Lobbyist Registration of National Insurance Crime Bureau, attached as Exhibits 81 and 82, (showing lobbyist registrations at the U.S. House and Senate).

[300] PSL § 114-a.

economic burden. Approving recovery for the RNG interconnection facilities would run afoul of the Commission's duties to ensure just and reasonable rates, to avoid disproportionate burdens on DACs, and to prioritize emissions reductions in DACs.[301]

Additionally, ratepayers should not finance a project that would contribute, directly or indirectly, to disproportionately burdening a DAC. As the Companies have noted, each of the four RNG production facilities will need to receive a Title V permit prior to operating. Under CLCPA Section 7(3), DEC will need to determine whether the permits would disproportionately burden DACs. The Commission should not require utility customers to foot the bill for the RNG interconnections before DEC has made a determination as to whether the RNG production itself is permissible under Section 7(3).

Q. Do you recommend approving recovery for costs related to the Newtown Creek RNG facility?

A. No. For the reasons discussed above, to the extent that RNG is produced, it should be reserved for truly hard-to-electrify uses such as certain industries and not used as a replacement for fossil gas in the existing distribution system.

Q. Do you recommend approving recovery for costs related to the HyGrid project?

A. No. Green hydrogen is a scarce resource and should be reserved for truly hard-to-electrify uses such as certain industries and not used as a replacement for fossil gas in the existing distribution system. Additionally, hydrogen blending—especially at levels higher than 5% by volume[302]—poses safety risks that would be inconsistent with the Commission's obligation to provide safe and reliable service.[303]

[301] *See* PSL § 65; CLCPA § 7(3).
[302] CPUC Report at 4.
[303] PSL § 65.

B. THE COMMISSION MUST ENSURE THAT AT LEAST 35–40% OF THE BENEFITS OF CLEAN ENERGY AND ENERGY EFFICIENCY SPENDING FLOW TO DACS.

Q. Do you have recommendations regarding the Companies' proposed spending and investments in DACs?

A. Yes. As discussed above, the Commission should require the Companies to develop a methodology to track, quantify, and report the health improvements from cleaner air, job creation, investment and wealth-building opportunities, business development opportunities, energy resilience, and utility bill reductions that result from the Companies' activities in DACs.

Additionally, the Commission should compare how the Companies' reported benefits in DACs measure compared to the 35–40% goal set by the CLCPA, and if the Companies are not meeting that goal, the Commission should require the Companies to develop a plan for reaching compliance with the CLCPA. Furthermore, the Companies should continue educating contractors with emerging technology and available state and federal rebates and incentives. The Companies should also expand their contractor partnerships to ensure they are working with a diverse workforce which speaks multiple languages.

The Companies must also scale up NPAs and thermal energy networks. Additionally, the Companies should propose additional job training and apprenticeship programs for thermal energy networks. I would also recommend that the Companies track and record the trainings they offered, the location of the trainings (times and days), enrollment, graduation rates, and hiring rates. Finally, because the Commission's Order

in the Thermal Energy Network Proceeding directed the utilities to develop one to five pilots, I would like to see each operating company commit to more than just the bare minimum and plan to pursue more than five pilots each.

C. RATEPAYERS SHOULD NOT PAY FOR POLITICAL ACTIVITIES

Q. Do you have other recommendations for adjustments to the rate request?

A. Yes. As discussed above, New York has adopted the Uniform System of Accounts, which requires that political advocacy expenditures be excluded.[304]

Under the Commission's 1977 policy statement ratepayers should not foot the bill for any "advertising which seeks to sway opinion — legislative, environmental, governmental, consumer or any other kind — to the industry's position on public policy disputes."[305] Additionally, only a very small percentage of a utility's operating revenues could go towards "institutional advertising," meaning advertising that is not clearly controversial or political, "for example, in which a company defends, justifies, or even merely explains and describes its activities."[306]

Q. What expenses should the Commission exclude as related to political activity?

A. The Commission should exclude expenses related to employee time dedicated to influencing public decisionmakers, such as legislative testimony and comments to the Climate Action Council, as these expenses must be recorded below the line under the Uniform System of Accounts.[307]

[304] 18 C.F.R. Pt. 201.
[305] *In re Advertising and Promotional Practices,* 1977 WL 201861 at *4.
[306] *Id*. at *5.
[307] 18 C.F.R. Pt. 201.

The Commission should also exclude expenses related to the Clean Energy Vision report, including expenses to produce and promote the report, from the Historic Test Year because National Grid uses the report in advertising to sway opinion on important public policy questions. The Commission should also exclude the costs of any other customer communications that promote natural gas. Because the Companies did not track individual expenses related to the report or customer communications, the Commission should require the Companies to produce a reasonable estimate of those expenses. The Commission should also clarify, going forward, that materials such as the Clean Energy Vision report that are used to promote National Grid's policy preferences and in lobbying and political advocacy must be recorded below the line.

To the extent the Commission disagrees that those expenses should be excluded altogether, at a minimum, the Clean Energy Vision report and customer communications that promote natural gas constitute "institutional advertising," and therefore, the Companies may only spend a small percentage of their operating revenues on such materials. Under either standard, the Companies should be required to track the time and money spent developing and proliferating these materials so that the Commission can ensure that ratepayers are not being improperly charged for them.

The Commission's 1977 Policy Statement explained that the Commission would determine the appropriate percentage for each utility, which would likely "range between 1/10 and 1/25 of one percent of operating revenues."[308] According to the Companies, their "prior rate plans have not specified a percentage or dollar amount that may be spent

[308] *In re Advertising and Promotional Practices*, 1977 WL 201861 at *5.

on 'institutional-type' advertising."[309] If this is true, then the Commission should set a percentage between 1/10 and 1/25 of operating revenues that the Companies may spend on institutional advertising in order to ensure compliance with the 1977 Policy Statement.

Additionally, in the 1977 Policy Statement, the Commission explained that "parties in rate cases would still be free to question the propriety of the lump sum allowance for the informational and other institutional advertising categories."[310] Therefore, after determining the appropriate percentage of operating revenues that the Companies may spend on institutional advertising, the Commission should allow parties the opportunity to provide input.

Finally, as discussed above, the Companies' request for recovery for dues paid to the National Insurance Crime Bureau must be denied pursuant to Section 114-a of the Public Service Law.

Q. Does this conclude your testimony?

A. Yes.

[309] Companies' Response to WE ACT Interrogatory 87(d), Ex. 56.
[310] *In re Advertising and Promotional Practices,* 1977 WL 201861 at *5.